

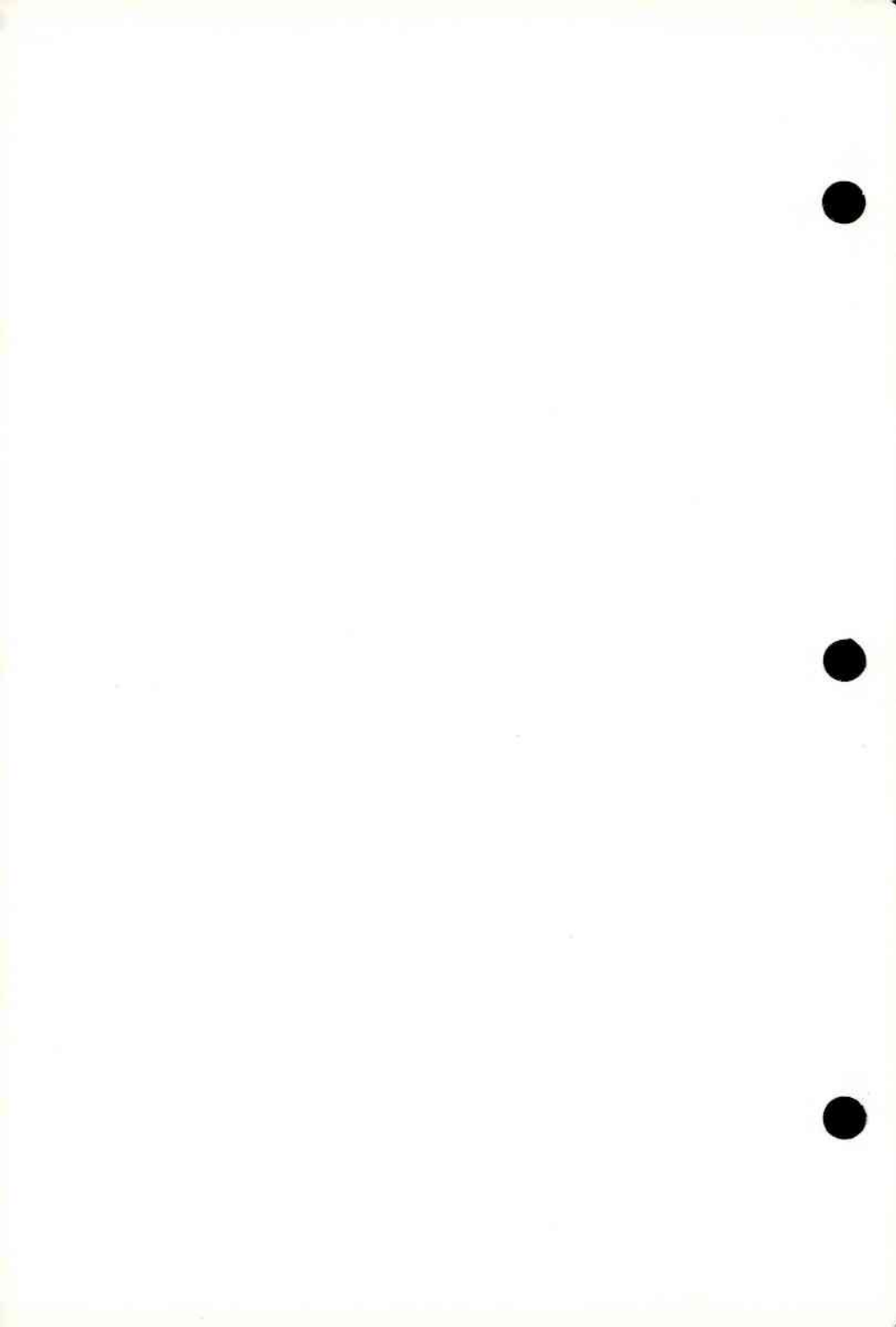
SL-1/2	
LAUNCH C/L SPT	
PART NO.	S/N
SKC32100150-801	1004

F2

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LIFTOFF CONFIGURATION

PANEL 1

EMS FUNC - ΔV
 EMS MODE - STBY
 GTA - off (down)
 EMS GTA COVER - Secure
 CMC ATT - IMU
 FDAI SCALE - 5/5
 FDAI SEL - 1/2
 FDAI SOURCE - CMC
 ATT SET - GDC
 MAN ATT ROLL - RATE CMD
 MAN ATT PITCH - ACCEL CMD
 MAN ATT YAW - RATE CMD
 LIM CYCLE - OFF
 ATT DBD - MIN
 RATE - HIGH
 TRANS CONTR PWR - on (up)
 RHC PWR NORM (2) - AC/DC
 RHC PWR DIR (2) - MNA/MNB
 SC CONT - SCS
 CMC MODE - FREE
 BMAG MODE ROLL - RATE 1
 BMAG MODE PITCH - RATE 1
 BMAG MODE YAW - RATE 1
 SPS THRUST - NORMAL (locked)
 ΔV THRUST (2) - OFF (guarded)
 SCS TVC PITCH - AUTO
 SCS TVC YAW - AUTO
 SPS GMBL MOT PITCH (2) - OFF
 SPS GMBL MOT YAW (2) - OFF
 ATV C GAIN - LO
 ELS LOGIC - OFF (guarded)
 ELS AUTO - MAN
 CM RCS LOGIC - on (up)
 CM PRPLNT DUMP - OFF (guarded)
 CM PRPLNT PURG - off (down) (guarded)
 IMU CAGE - off (down) (guarded)

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EMS ROLL - OFF
.05G sw - OFF
Pc IND sw - Pc
LV/SPS IND sw - SIVB
TVC GMBL DR PITCH - AUTO
TVC GMBL DR YAW - AUTO
EVNT TMR RSET - up (ctr)
EVNT TMR STRT - ctr
EVNT TMR MIN - ctr
EVNT TMR SEC - ctr

PANEL 2

PL VENT vlv - push (lock)
DOCK PROBE EXTD/REL - OFF (guarded)
DOCK PROBE EXTD/RETR (2) tb - gray
DOCK PROBE RETR PRIM - OFF
DOCK PROBE RETR SEC - OFF
S5 - N/A down
SPOT LIGHT - off (ctr)
TUNL LT - OFF
MSN TMR - START
SM RCS PSM 1 He - ctr (CLOSE*)
SM RCS PSM 1 He tb - bp
SM RCS PSM 1 MANF ISOL - ctr (OPEN*)
SM RCS PSM 1 MANF ISOL tb - gray
UP TLM CM - BLOCK
CM RCS PRESS - off (down) (guarded)
RCS IND sw - TK PRESS/QTY
SM RCS QUAD He (4) - ctr (OPEN*)
SM RCS QUAD He (4) tb - gray
SM RCS PSM PRPLNT (4) - ctr (CLOSE*)
SM RCS PSM PRPLNT (4) tb - bp
SM RCS QUAD PRPLNT (4) - ctr (OPEN*)
SM RCS QUAD PRIM PRPLNT (4) tb - gray
SM RCS QUAD SEC PRPLNT (4) tb - gray
RCS CMD - ctr (OFF*)
RCS TRNFR - ctr (SM*)
CM RCS PRPLNT (2) - ctr (on,up*)
CM RCS PRPLNT tb (2) - gray
SM RCS SEC FUEL PRESS (4) - ctr (CLOSE*)

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EDS AUTO - on (up)
DOCK RING SEP (2) - off (down) (guarded)
CM/SM SEP (2) - off (down) (guarded)
PRPLNT DUMP - AUTO
2 ENG OUT - AUTO
LV RATES - AUTO
TWR JETT (2) - AUTO (down) (guarded)
LV GUID - IU
MN REL - off(down)(guarded)
MSN TMR HR, MIN, SEC - ctr
C/W NORM - BOOST
C/W CSM - CSM
C/W PWR - 1
C/W LAMP TEST - off (ctr)
C/W MEMORY - ctr (RSET*)
RCS IND sel - SM D
CAB FANS (2) - OFF
H2 HTRS (2) - AUTO
O2 PRESS IND - SURGE TK
O2 HTRS (2) - AUTO
H2 FANS (2) - AUTO
ECS IND sel - PRIM
ECS RAD FLOW AUTO CONT - AUTO
ECS RAD tb - gray
ECS RAD FLOW PWR CONT - off (ctr)
ECS RAD MAN SEL - RAD 1
ECS RAD PRIM HTR - off (ctr)
ECS RAD SEC HTR - OFF
POT H2O HTR - OFF
SUIT CKT H2O ACCUM AUTO - 1
SUIT CKT H2O ACCUM ON - ctr
SUIT CKT HT EXCH - off (ctr) (ON*)
SEC COOL LOOP EVAP - off (ctr) (RSET*)
SEC COOL LOOP PUMP - off (ctr)
SEC EVAP H2O FLOW - off (ctr)
GLY EVAP IN TEMP - MAN
GLY EVAP STM PRESS AUTO - MAN
GLY EVAP STM PRESS INCR - ctr (vlv partially open)
GLY EVAP H2O FLOW - off (ctr)
H2O QTY IND sw - POT
SM H2O TK INLET - CLOSE

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PANEL 3

VHF ANT - SM LEFT
 SPS ENG INJ VLV ind (4) - CLOSE
 FC RAD (2) - off (ctr) (NORM*)
 FC RAD (2) tb - gray
 FC HTRS (2) - on (up)
 FC IND sel - 1
 SPS QTY TEST - off (ctr)
 OXID FLOW VLV INCR - NORM
 OXID FLOW VLV PRIM - PRIM
 PUG MODE - NORM
 FC PURG (2) - OFF
 SM PWR SOURCE 1 MNA - ctr (on,up*)
 SM PWR SOURCE 1 MNA tb - gray
 SM PWR SOURCE 2 MNA - OFF
 SM PWR SOURCE 2 MNA tb - bp
 SM PWR SOURCE 3 MNA - OFF
 SM PWR SOURCE 3 MNA tb - bp
 MNA RSET - ctr (RSET*)
 SPS He vlv (2) - AUTO
 SPS He vlv tb (2) - bp
 SPS TEMP IND sw - OXID LN
 SPS PRESS IND sw - He
 FC REACS (2) - ctr (on,up*)
 FC REACS (2) tb - gray
 H2 VENT - ctr (OFF*)
 H2 VENT tb - gray
 SM PWR SOURCE 1 MNB - OFF
 SM PWR SOURCE 1 MNB tb - bp
 SM PWR SOURCE 2 MNB - OFF
 SM PWR SOURCE 2 MNB tb - bp
 SM PWR SOURCE 3 MNB - ctr (on,up*)
 SM PWR SOURCE 3 MNB tb - gray
 MNB RSET - ctr (RSET*)
 DC IND sel - MNA
 BAT CHARGE - OFF
 S BD XPNDR - PRIM
 S BD PWR AMPL PRIM - PRIM
 S BD PWR AMPL HI - HIGH
 PWR AMPL tb - gray

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S BD MODE VOICE - VOICE
 S BD MODE PCM - PCM
 S BD MODE RNG - RNG
 S BD AUX TAPE - ctr
 S BD AUX TV - ctr
 UP TLM DATA - DATA
 UP TLM CMD - NORM
 S BD ANT OMNI - B
 S BD ANT - OMNI
 VHF AM SQUELCH A tw - noise +1
 VHF AM A - off (ctr)
 VHF AM B - DUPLEX
 VHF AM RCV - off (ctr)
 VHF BCN - OFF
 VHF RNG - OFF
 S BD SQUELCH - ENABLE
 FC REACS vlv - LATCH
 H2 PURG LINE HTR - OFF
 TV SOURCE - CM
 VHF AM SQUELCH B tw - noise +1
 TAPE RCDR PCM - PCM/ANLG
 TAPE RCDR RCD - RCD
 TAPE RCDR FWD - FWD
 TAPE MOTION tb - gray
 SCE PWR - NORM
 PMP PWR - NORM
 PCM BIT RATE - HIGH
 PTT BU - NORM
 AC INV 1 - MNA
 AC INV 2 - MNB
 AC INV 3 - OFF
 INV 1 AC 1 - on (up)
 INV 2 AC 1 - OFF
 INV 3 AC 1 - OFF
 AC 1 RSET - ctr (RSET*)
 INV 1 AC 2 - OFF
 INV 2 AC 2 - on (up)
 INV 3 AC 2 - OFF
 AC 2 RSET - ctr (RSET*)
 AC IND sel - BUS 2 ØC

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PANEL 4

SPS GAUGING - OFF
TELCOM GRP 1 - AC1
TELCOM GRP 2 - AC2
GLY PUMPS - 1 AC1
SUIT COMPR 1 - AC1
SUIT COMPR 2 - OFF
cb Panel 4 - all closed

PANEL 5

MN BUS TIE (2) - on (up)
SM PWR SOURCE 1 - FC1
SM PWR SOURCE 3 - FC3
FC1 PUMPS - AC1
FC3 PUMPS - AC2
SM RCS ENG PKG HTRS (4) - off (ctr)
SM RCS QUAD HTRS (4) - off (ctr)
SM RCS PSM 1 HTRS - off (ctr)
SPS HTRS - OFF
INTGL LTS - as desired
FLOOD LTS - OFF, (full dim or full brt)
FLOOD DIM - 1
FLOOD FIXED - OFF
BAT CHARGER - AC1
cb Panel 5 all closed except:
cb FLT/PL BUS BAT A,B,C, (3) - open
~~cb SEC COOL HTR CONTR MNA - open~~
cb H2O/UR DUMP HTR (2) - open
cb 02 VAC ION PUMPS (2) - open

PANEL 6

MODE - INTERCOM/PTT
PWR - AUDIO/TONE
PAD COMM - OFF
INTERCOM - T/R
S BD - T/R
VHF AM - T/R
AUDIO CONT - NORM
SUIT PWR - on (up)
tw settings - as desired

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PANEL 7

EDS PWR - on (up)
SCS TVC SERVO PWR #1 - AC1/MNA
SCS TVC SERVO PWR #2 - AC2/MNB
FDAI/GPI PWR - BOTH
LOGIC 2/3 PWR - on (up)
SCS ELEC PWR - GDC/ECA
SCS SIG CONDR/DR BIAS 1 - AC1
SCS SIG CONDR/DR BIAS 2 - AC2
BMAG PWR (2) - ON
DIRECT 02 vlv - OPEN (CCW) (>2 in. H2O on SUIT/CAB Δ P ind)
(02 flow - 0.7-0.9 lb/hr)

PANEL 8

cb Panel 8 - all closed except:

- cb CM RCS HTRS (2) - open
- cb SPS GAUGING (4) - open
- cb FLOAT BAG (3) - open

AUTO RCS SEL A/C ROLL A1 - OFF
AUTO RCS SEL A/C ROLL C1 - OFF
AUTO RCS SEL A/C ROLL A2 - OFF
AUTO RCS SEL A/C ROLL C2 - OFF
AUTO RCS SEL B/D ROLL B1 - MNA
AUTO RCS SEL B/D ROLL D1 - MNB
AUTO RCS SEL B/D ROLL B2 - MNA
AUTO RCS SEL B/D ROLL D2 - MNB
AUTO RCS SEL PITCH A3 - MNB
AUTO RCS SEL PITCH C3 - MNA
AUTO RCS SEL PITCH A4 - MNA
AUTO RCS SEL PITCH C4 - MNB
AUTO RCS SEL YAW B3 - MNA
AUTO RCS SEL YAW D3 - MNB
AUTO RCS SEL YAW B4 - MNB
AUTO RCS SEL YAW D4 - MNA

INT NUM LT - as desired

INT INTGL LT - as desired

INT FLOOD LT - OFF, full dim, or full brt

FLOOD LTS DIM - 1

FLOOD LTS FIXED - OFF

FLOAT BAG (3) - VENT (locked)

SECS LOGIC (2) - on (up) (locked)

SECS PYRO ARM (2) - on (up) (locked)

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PANEL 9

MODE - INTERCOM/PTT
PWR - AUDIO/TONE
PAD COMM - OFF
INTERCOM - T/R
S BD - T/R
VHF AM - T/R
AUDIO CONT - NORM
SUIT PWR - on (up)
VHF RNG - NORM
tw settings - as desired

PANEL 10

MODE - INTERCOM/PTT
PWR - AUDIO/TONE
PAD COMM - OFF
INTERCOM - T/R
S BD - T/R
VHF AM - T/R
AUDIO CONT - NORM
SUIT PWR - on (up)
tw settings - as desired

PANEL 12

MDA TUNL VENT vlv - MDA/CM Δ P

PANEL 13

FDAI sw (2) - INRTL
EARTH/LUNAR - PWR OFF
ALT SET - 104
LTG - OFF
MODE - HOLD/FAST
SLEW - off (ctr)

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PANEL 15

COAS PWR - OFF
UTIL PWR - OFF
PL BCN LT - off (ctr)
PL DYE MARKER - off (down)(guarded)
PL VENT - OFF

PANEL 16

e-p SPECT - ON
UTIL PWR - OFF

PANEL 98

XMIT/ICOM - off (ctr)
CALL/SLEEP - SLEEP
VOL tw - as req'd
SPKR/HEADSET - HEADSET

PANEL 100

UTIL PWR - ON (CSM 116 only)
FLOOD LTS DIM - 1
FLOOD LTS FIXED - OFF
G/N POWER/OPTICS - OFF
G/N POWER/IMU - on (up)(guarded)
G/N LTS - AC1
NUMERICS LT - as desired
FLOOD LTS - off, full dim, or full bright
INTGL LT - as desired

PANEL 101

SYS TEST (LH) - 3
SYS TEST (RH) - B
CM RCS HTRS - OFF
WASTE H2O DUMP - HTR A
UR DUMP - HTR A

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PANEL 122

OPT ZERO - ZERO
OPT TELTRUN - SLAVE TO SXT
OPT COUPLING - DIRECT
OPT MODE - MAN
OPT SPEED - LO
COND LAMPS - ON
UP TLM - ACCEPT

PANEL 163

GLYCOL EVAP TEMP IN - NORM

PANEL 164

S1 - ON (verified at panel closeout)

PANEL 201

C/W INPUT (all) - ENABLE

PANEL 223

R C/W TONE ADJUST - as des'd
L C/W TONE ADJUST - as des'd
CTR C/W TONE ADJUST - as des'd

PANEL 225

cb Panel 225 - all closed

PANEL 226

cb Panel 226 - all closed except:
cb FC1 RAD BAT RLY - open
cb H2 VENT BAT RLY - open
cb FC3 RAD BAT RLY - open
cb COAS/TUNL LTG MNB - open
cb CB40 - open

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PANEL 227

SCI PWR - on (up) (CSM 116 only)

PANEL 229

cb Panel 229 all closed except:
cb PYRO BUS A BAT BUS A - open
cb PYRO BUS B BAT BUS B - open
cb MAIN REL PYRO (2)- open

PANEL 230

cb Panel 230 - all open
CSM/SWS INTERFACE PWR (2) - DISCONNECT
CSM/SWS INTERFACE PWR tb - gray
CSM/SWS INTERFACE SIG - DISCONNECT
CSM/SWS INTERFACE SIG tb - gray

PANEL 250

cb Panel 250 - all closed except:
cb MN BUS INTERCONNECT (2) - open

PANEL 251

WASTE MGMT OVBD DRAIN vlv - OFF

PANEL 252

BAT VENT vlv - VENT
WASTE STOWAGE VENT vlv - VENT

PANEL 274

cb Panel 274 - all closed

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PANEL 275

cb Panel 275 - all closed except:
cb MNA BAT C - open
cb MNB BAT C - open
cb BAT BUS A MNA - open
cb BAT BUS A BAT C - open
cb BAT BUS A PYRO BAT A - open
cb BAT BUS B MNB - open
cb BAT BUS B BAT C - open
cb BAT BUS B PYRO BAT B - open

PANEL 276

cb Panel 276 - all closed

PANEL 277 (CSM 117 only)

EXP S071 - ON
EXP S072 PWR - on (up)
EXP S072 START - OFF
EXP DATA - REAL TIME

PANEL 278

cb UPRT SYS COMPR (2) - open
cb DOCK RING SEP (2) - open

PANEL 300

RH SUIT FLOW v1v - FULL FLOW

PANEL 301

LH SUIT FLOW v1v - FULL FLOW

PANEL 302

CTR SUIT FLOW v1v - FULL FLOW

PANEL 303

PRIM CAB TEMP v1v - COLD (CW)
SEC CAB TEMP v1v - MAX COOL (CW)

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PANEL 304

DRNK H2O SUPPLY vlv - OFF (CW)

PANEL 305

FOOD PREP COLD H2O vlv - rel

FOOD PREP HOT H2O vlv - rel

PANEL 306

MSN TMR - START

EVNT TMR RSET - UP (ctr)

EVNT TMR STRT - ctr

EVNT TMR MIN - ctr

EVNT TMR SEC - ctr

MSN TMR HR - ctr

MSN TMR MIN - ctr

MSN TMR SEC - ctr

PANEL 325

CAB PRESS RELF vlv (RH) - BOOST/ENTRY

CAB PRESS RELF vlv (LH) - BOOST/ENTRY

PRIM GLY TO RAD vlv - BYPASS (pull)

PANEL 326

REPRESS PKG vlv - ON

SM 02 SUPPLY vlv - ON

SURGE TK 02 vlv - ON

GLY RSVR IN vlv - OPEN

GLY RSVR BYPASS vlv - CLOSE

GLY RSVR OUT vlv - OPEN

PANEL 350

C02 CSTR DIVERT vlv - both (ctr)

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PANEL 351

MAIN REG vlv (2) - OPEN
H2O/GLY TK PRESS REG vlv - BOTH
H2O/GLY TK PRESS RELF vlv - BOTH
EMER CAB PRESS vlv - OFF
CAB REPRESS vlv - OFF (CCW)

PANEL 352

WASTE TK SERVICING vlv - CLOSE
PRESS RELF vlv - RELF
POT TK IN vlv - ~~CLOSE~~ as req'd at closeout
WASTE TK IN vlv - AUTO

PANEL 375

SURGE TK PRESS RELF vlv - OPEN (CW)

PANEL 376

PLVC - NORMAL (up)

PANEL 377

GLY TO RAD SEC vlv - BYPASS (CCW)

PANEL 378

PRIM GLY ACCUM vlv - open (CCW)

PANEL 379

PRIM ACCUM FILL vlv - OFF (CW)

PANEL 380

O2 DEMAND REG vlv - BOTH
SUIT TEST vlv - OFF
SUIT CKT RET vlv - CLOSE (push)

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PANEL 382

SUIT HT EXCH PRIM GLY vlv - FLOW (CCW)
SUIT FLOW RELF vlv - OFF
PRIM GLY EVAP IN TEMP vlv - MIN (CCW)
SUIT HT EXCH SEC GLY vlv - FLOW (CCW)
SEC EVAP H2O CONT vlv - AUTO (CW)
PRIM EVAP H2O CONT vlv - AUTO (CW)
H2O ACCUM vlv (2) - RMTE (CCW)

PANEL 399

AUX GLY EVAP IN TEMP vlv - MIN (CW)

PANEL 600

EMER 02 vlv - CLOSE

PANEL 601

REPRESS 02 vlv - CLOSE (guarded)

PANEL 602

REPRESS 02 RELF vlv - OPEN (CW)

PANEL 603

IVA STA 02 SUP - OFF

PANEL 604

IVA PWR - OFF

PANEL S015 (EXP)

40X CAMR - NORM

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FWD HATCH

PRESS EQUAL vlv - CLOSE
ACTR HNDL sel - stow/check locked

SIDE HATCH

CAB PRESS DUMP vlv - CLOSE (CW)
GEAR BOX sel - LATCH
ACTR HANDLE sel - UNLATCH
LOCK PIN REL KNOB - LOCK
LOCK PIN ind - flush
GN2 VLV HANDLE - outboard
BPC JETT KNOB - toward BPC JETT

* - last momentary position before liftoff.

NC1 TARGETING PAD

		NOMINAL						NOMINAL UPDATE						PRELAUNCH UPDATE					
N95	HR	+	X	X	X	0	0	2	+	X	X	X	0	0	2	+	X	X	X
TIG NC1	MIN	+	X	X	X	X	2	0	+	X	X	X	2	3		+	X	X	X
	SEC	+	X	X	1	1	9	0	+	X	3	0	0	0		+	X	X	X
N57 HALF REVS			X	X	X	X	0	3		X	X	X	0	3			X	X	X
N37	HR	+	X	X	X	0	0	6	+	X	X	X	0	0	7	+	X	X	X
TIG TPI	MIN	+	X	X	X	X	4	9	+	X	X	X	0	2		+	X	X	X
	SEC	+	X	X	3	8	7	0	+	X	5	5	0	0		+	X	X	X

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TALIGN PAD

		NOMINAL						NOMINAL UPDATE						PRELAUNCH UPDATE					
N34	HR	+	X	X	X	0	0	6	+	X	X	X	0	0	6	+	X	X	X
GET	MIN	+	X	X	X	X	3	9	+	X	X	X	5	5		+	X	X	X
ALIGN	SEC	+	X	X	5	2	6	0	+	X	1	5	0	0		+	X	X	X

BOOST PREPARATION

-25:00

Change X STABLE MEMBER AZIMUTH, if necessary:

```
*V78E *
*F 06 29 X SM AZ (.01° *
*V21E *
*Load new Azimuth +0 *
*PRO *
*ALIGN GDC *
```

AUTO RCS A/C ROLL (4) - OFF (verify)
 AUTO RCS B/D ROLL B1 & B2 - MNA
 AUTO RCS B/D ROLL D1 & D2 - MNB
 AUTO RCS PITCH A3 & C4 - MNB
 AUTO RCS PITCH C3 & A4 - MNA
 AUTO RCS YAW B3 & D4 - MNA
 AUTO RCS YAW D3 & B4 - MNB

-15:00

CTE UPDATE VERIFICATION

DC IND sel - BAT C
 DC VOLTS ind - 35-37.5 vdc
 DC IND sel - MNA
 FDAI-1 total att R=90+AZ, P=90, Y=0
 FDAI SCALE - 5/5
 RATE - HIGH
 TRANS CONTR PWR - on(up) (verify)
 RHC PWR DIRECT(2) - MNA/MNB
 CMC MODE - FREE
 BMAG MODE (3) - RATE 1
 RHC #2 - ARMED

ASTRO LAUNCH OPERATIONS VOICE CHECK

PLT S BD sw - OFF
 CDR VHF AM sw - OFF

VOICE CHECK WITH MCCH

PLT S BD sw - T/R
 CDR VHF AM sw - T/R
 SPS THRUST - NORMAL (locked)
 ΔV THRUST (2) - OFF

ARN.

BOOST PREPARATION

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DATE

EDS AUTO - on (up)
2 ENG OUT - AUTO
LV RATES - AUTO
RCS CMD - OFF
TVC SERVO PWR #1 - AC1/MNA
TVC SERVO PWR #2 - AC2/MNB

-10:00 FC REAC vlv - LATCH

-08:30 SEC COOL LOOP PUMP - off (ctr) (verify)

-04:10 L/V ENGINE lts (8) - on

-04:00 ASTRO LAUNCH OPERATIONS COMM CHECK

DSKY - Verify P02
V75 (Do not ENTR)
TAPE RCD FWD - FWD (tb-gray)

-2:15 PRIM GLY TO RAD - pull (bypass)

-1:15 MN BUS TIE (2) - on (up)

-1:00 PAD COMM (2) - OFF
VHF AM VOL tw - increase to above
normal listening level

-00:45 GDC ALIGN pb - PUSH & HOLD
R=90+AZ, P=90, Y=0
FDAI 2 Total att - no motion
GDC ALIGN pb - release

BOOST PREPARATION

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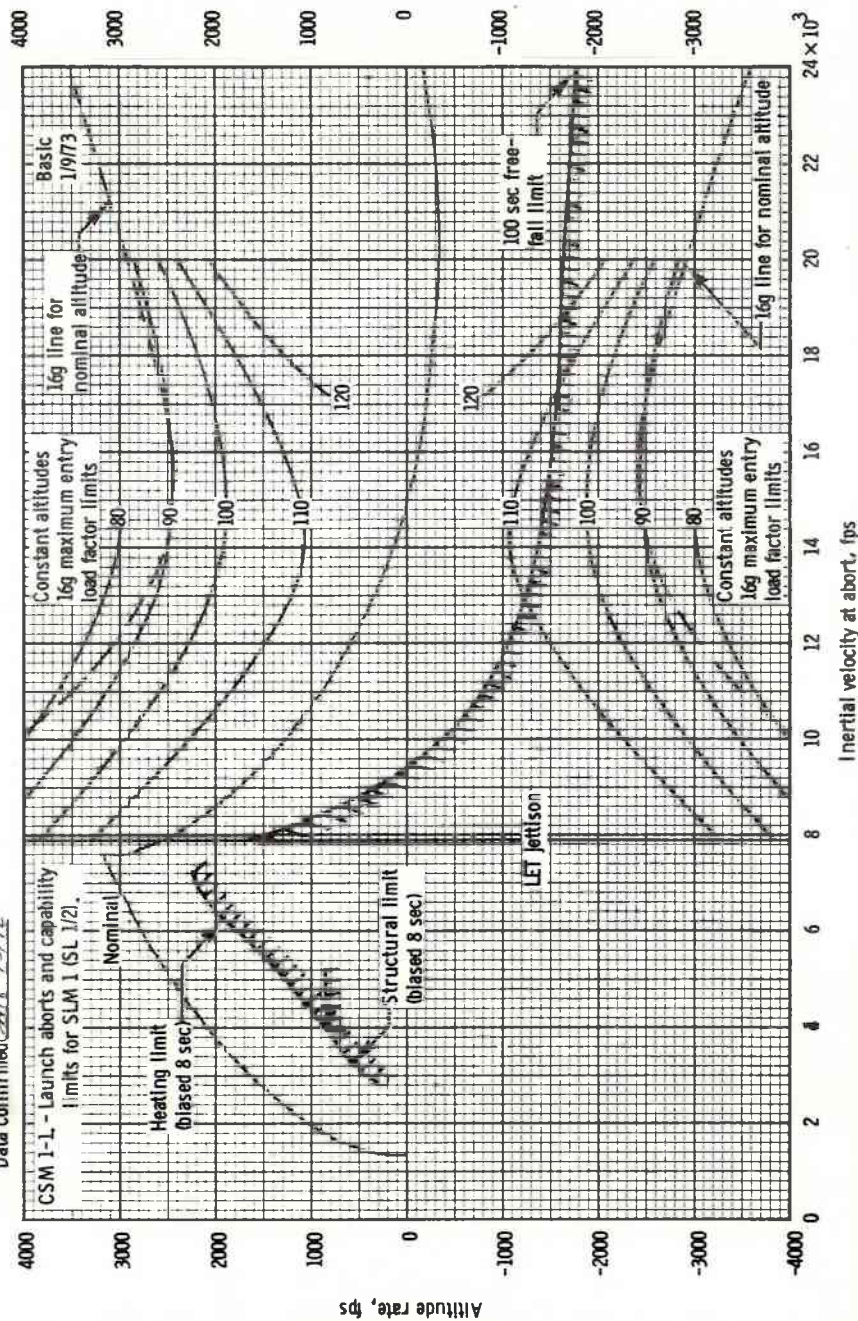
DATE 2/5/73

Henderson/FB/MPAD

SL-2(SA-206) LV Operational Traj. (MSC Tape #A04529)

Data source SL 003 Vol. II Rev. B Oct. 1972 & Vol. III Amend. #106 Nov. 3, 1972

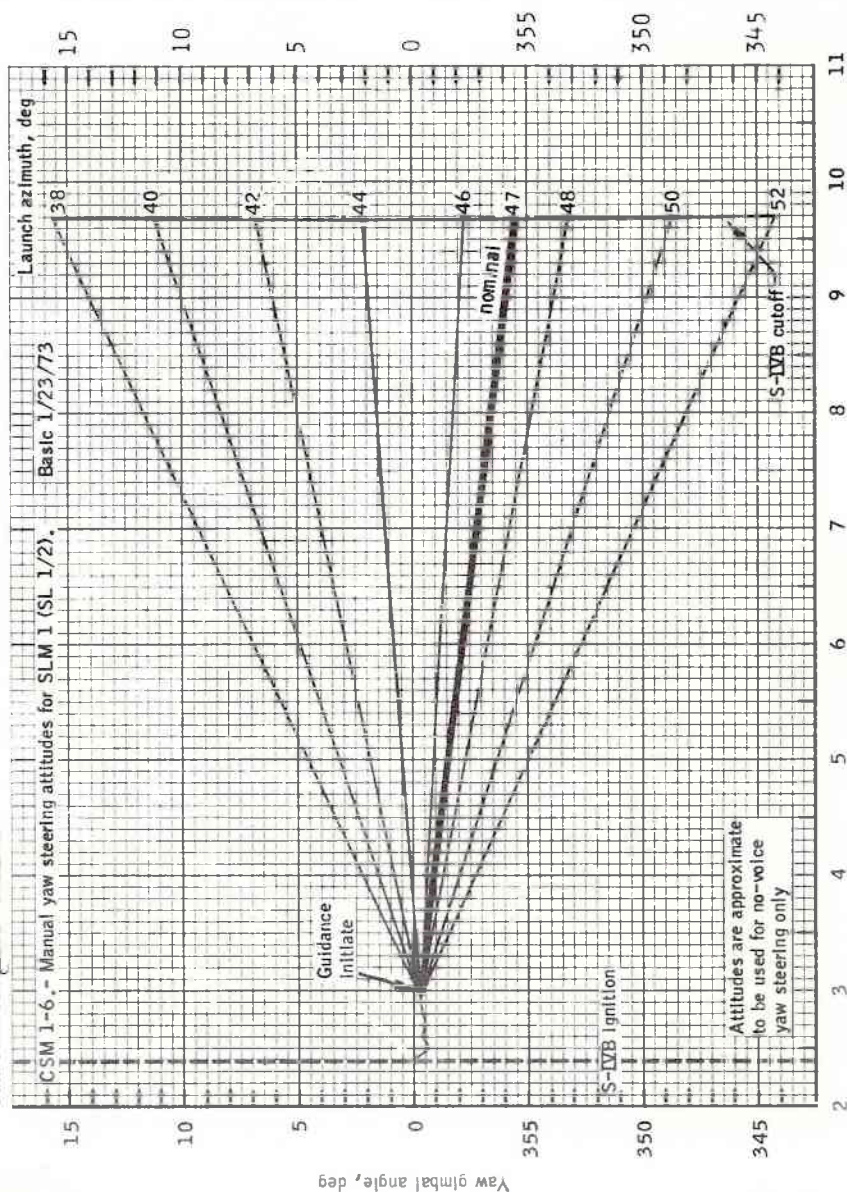
Data confirmed 1/15/72



LAUNCH ABORT
LIMITS

YAW STEERING ATTITUDES

Henderson/FB/MPAD
Data source MSEC/SAFE-AERO-MFP
Data confirmed JAH 1/23/73



L
2-4

Ground elapsed time, min

DATE 4/6/73

MANUAL STEERING CREW CHART
(HP + HA = 274 nm)

CURRENT ALTITUDE (H, nm)	HP/HA (nm)	INERTIAL VELOCITY (VI, fps)
70	70/204	25 930
72	72/202	25 910
74	74/200	25 900
76	76/198	25 880
78	78/196	25 870
80	80/194	25 850
82	82/192	25 840
84	84/190	25 820
86	86/188	25 810
88	88/186	25 800
90	90/184	25 780
92	92/182	25 770
94	94/180	25 750
96	96/178	25 740
98	98/176	25 720
100	100/174	25 710

NOTE: Assumes inserting at HDOT = 0.
SIVB acceleration at insertion = $\sim 83 \text{ ft/sec}^2$.
CSM acceleration at insertion = $\sim 25 \text{ ft/sec}^2$.

ALTITUDE vs VI

DATE 5/22/73

BOOST

BOOST

-00:03 Ignition CMD
-00:01 L/V ENGINES lts (8) - out
00:00 LIFTOFF lt - on

00:00

*After LIFTOFF verified: *
* If LIFTOFF lt off: - push *
* If NO AUTO ABORT lt on: - push*

Clock Start (auto) - report
MET Resets & starts counting up auto
P11 auto

+5°/sec P,Y
+20°/sec R

*If no P11: - Key ENTR *
* Start DET & reset MET*

06 62 VI,H DOT, H PAD (fps,fps,.1nm)
*If LV GUID lt on: *
* LV GUID - CMC *

MODE IA

+00:10 Roll & Pitch Program - report

If LOX TK PRESS >50 psia:
* Abort immediately *

CABIN RELIEVING ~14K(2.3 nm) - report

*If no Press decrease ~25K(4.1 nm): *
* CAB PRESS RELIEF vlv(RH)-DUMP *

+00:56 Roll complete - report

+01:01 MODE IB - report
PRPLNT DUMP - RCS CMD

01:01

+01:14 MAX Q
V82E, N62E

+5°/sec P,Y
+20°/sec R

+01:40 EDS AUTO - OFF - report
2 ENG OUT - OFF
LV RATES - OFF

MODE IB

DATE 2/5/73

+01:48 MODE IC - report

+02:11 CMC Boost Polynomial stops

GO/NO GO FOR STAGING - report+02:18 INBOARD CUTOFF - (1ts 5,6,7,8 on)
LIFTOFF 1t - out+02:21 OUTBOARD CUTOFF - report (1ts on)+02:23 SIB STAGING (1ts out)+02:24 SIVB Ign Command (1t on)+02:27 SIVB 65% (1t out)+02:51 TWR JETT (2) - on (up) (TFF>1+20)
(OECD + 30 sec)

*NO TWR JETT, pg L/4-2 *

Twr Jett - report

V46E, V46E, N62E

MAN ATT PITCH - RATE CMD

GLY EVAP STEAM PRESS - AUTO

GLY EVAP H2O FLOW - AUTO

+02:52 Guidance Initiate - report+03:22 Guidance Good+04:00 Report Status+05:00 Report Status+06:00 Report Status+06:15 OMNI ANT - ~~CH~~ C+07:00 Report StatusGMBL MOT (4) - START - ON (PLT Confirm)

Check GPI:

LV/SPS IND - GPI (Momentarily)

PITCH = ~~+1.2°~~, YAW = +0.4°

+1.1°

+07:49 PU SHIFT

H=16.5 nm

+5°/sec, P&Y
+20°/sec, R

02:18

+10°/sec, P&Y
+20°/sec, R

MODE IC

TWR JETT

MODE II

+10°/sec P,Y
+20°/sec R

2/5/73

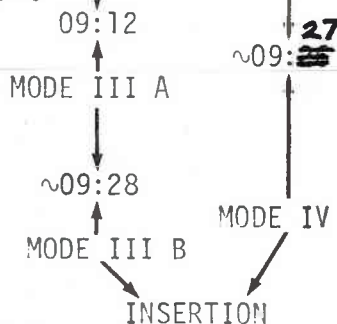
DATE

4/30/73 (pat)

L
2-9

+08:00 Report Status
+09:00 GO/NO GO FOR ORBIT - report

+09: ~~26~~²⁷ Mode IV - Report
(VI ~ 24,470)
(H DOT ~ -145)
(H ~ 84)



+09: ~~12~~¹³ SECO (1t on) - report _____

*If LV GUID - CMC: *
* THC - CCW *
* & neutral in 1 sec *

*If no SECO, (at VI): *
* THC - CCW *
* & neutral in 1 sec *

Log VI _____ (fps)
H DOT _____ (fps)
H PAD _____ (.1nm)

KEY RLSE

Log HA _____ (.1nm)
HP _____ (.1nm)
TFF _____ (min-sec)

PRO

V37E 00E

Verify LOX TK PRESS decreasing to ~32 psia
and FUEL TK PRESS decreasing

*If $\Delta P > 36$ psid (OXID > FUEL) or *
*If $\Delta P > 26$ psid (FUEL > OXID) or *
*If LOX or FUEL TK PRESS > 50 psia: *
* EMERGENCY CSM/LV SEP, pg EMER/1-1 *

DATE 4/6/73
5/22/73 (P:4)

L
2-10

GMBL MTRS (4) - OFF (PLT confirm)
CM RCS LOGIC - OFF
cb ELS/CM-SM SEP (2) - open
cb FLT/PL VENT - open
MN BUS TIE (2) - OFF
CAB PRESS REL vlv (2) - NORMAL/LATCHED
PCM BIT RATE - LOW
~~Verify DSE tape motion~~
~~(LBR/RCD/FWD/CMD RESET)~~
VHF AM A - SIMPLEX
VHF AM B - OFF
~~5 BD AUX TAPE - TAPE~~

CSM/LV SEPARATION

*If LV GUID - CMC: *
* Do not reload DAP *
* Mnvr to SEP ATT (180°, 317°, 0°)*

Load RCS DAP
R1=11103, R2=01111
V46E
Load N22 (180°, 170°, 2°), V62E
SM RCS QUAD PRPLNT tb (8) - gray (verify)
AUTO RCS SELECT (12) - MNA/MNB (verify)
EMS FUNC - OFF
Set ΔVC to -100.0
EMS FUNC - ΔV
FDAI SCALE - 5/1
ATT DB - MIN/LOW
SC CONT - CMC
BMAG MODE (3) - RATE 2

GO/NO-GO for CSM/LV SEP

14:00 V37E 47E F 16 83 ΔV X,Y,Z (.1fps)
THC - ARMED
RHC #2 - ARMED (verify)
RCS CMD - ON

*If LV GUID - CMC: *
* Insure rates nulled *
* Load DAP 11103, 01111 *
* V46E, V62E *

15:30 EMS MODE - NORMAL

5/22/73 (P&I)
1/22/73
#16473

DATE

15:58 Thrust +X and hold
15:59 CMC MODE - AUTO
16:00 CSM/LV SEP pb - push, hold, and release
LV TANK PRESS - full scale low

*If No Separation: *
* THC - CCW (4 sec min) *
* DET reset and counting up (auto) *
* LV TK PRESS - full scale low (SEP ind)*

~16:07 THC - release ($\Delta V = 3.0$ fps)
PRO

F 37 OOE

V49E (Mnvr to 180°, 170°, 2° and observe SLA Panels)

POST SEPARATION

SM RCS QUAD PRPLNT tb (8) - gray (verify)
SM RCS QUAD He tb (4) - gray (verify)
SM RCS QUAD SEC FUEL PRESS (4) - OPEN
FC REAC vlv - NORM
SECS PYRO ARM (2) - SAFE
SECS LOGIC (2) - OFF
cb SECS ARM (2) - open
EDS PWR - OFF
cb EDS (3) - open
THC & RHC - LOCKED
EMS - OFF/STBY
ATT DB - MIN/HIGH
THC PWR - OFF
RHC PWR DIRECT (2) - OFF
LV/SPS IND - GPI
TVC SERVO PWR (2) - OFF
cb RCS LOGIC (2) - open
cb DIRECT ULLAGE (2) - open

(326) REPRESS PKG vlv - OFF

INSTALL COAS

SM RCS ENG PKG HTRS (4) - 1

SM RCS QUAD HTRS (4) - PRIM

SM RCS PSM 1 HTRS - PRIM

SPS HTRS - PRIM

C/W - NORMAL

BPC JETT KNOB - 180° from BPC JETT

GN2 vlv HNDL - VENT (pull)

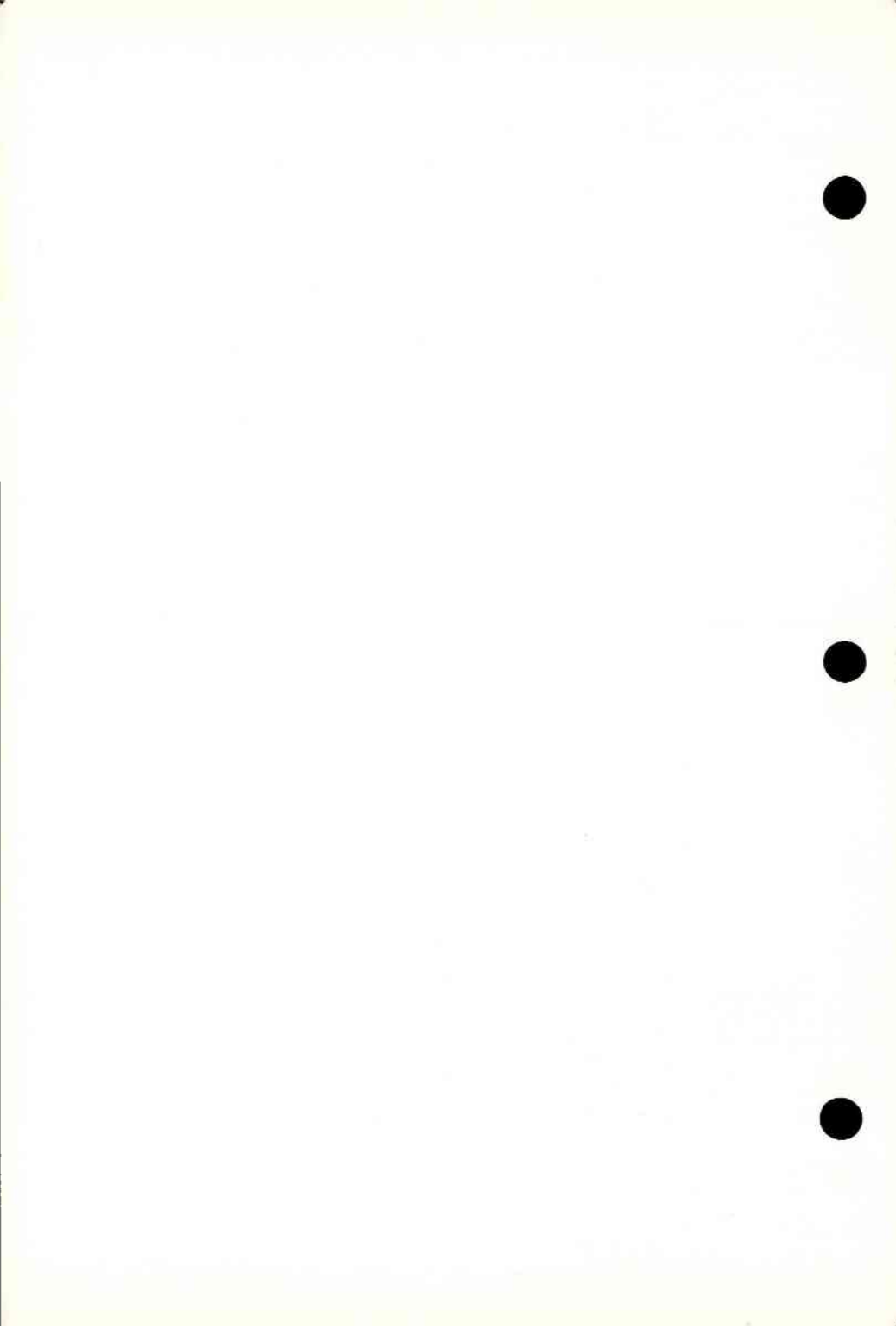
HATCH GEAR BOX - LATCH (verify)

ACTR HNDL SELECTOR - neutral

Verify cabin press > 4.7 psia
and O₂ flow not pegged hi.
Side Hatch Dump vlv - OPEN
(Until cabin press = 4.0 psia)
DIRECT O₂ vlv - OPEN (ccw)
(Until cabin press = 5.7 psia)

4/6/73
5/22/73 (RET)

DATE



SYSTEMS CHECKS

1 CONFIRM NORMAL SUIT AND CABIN PRESSURE

Verify cabin press >4.7 psia

~~and O2 flow not pegged hi~~

(351) EMERG CABIN PRESS vlv - BOTH

(380) SUIT CKT RET vlv - open (pull)

Remove helmets & gloves

Stow helmets in helmet bags (U2), gloves in accessory bags (U2) & put in PGA bag (U2)

2 SPT to LEB

STRUT UNLOCK LANYARD (2) - STOW

(304) DRINKING WATER SUPPLY vlv - ON (CCW)

cb COAS/TUNL LTG MNB - close

Unstow Tool E (L2)

3 CDR & SPT go to Rendezvous book
PLT continue with Systems Checks

NOTE: The following steps must be performed prior to NC1 and may be rearranged as req'd for each mission.

4 MAIN REG CHECK

(351) MAIN REG B vlv - close

EMER CABIN PRESS sel - 1

PUSH TO TEST PB - PUSH (O2 FLOW INC)

MAIN REG B vlv - open

MAIN REG A vlv - close

EMER CABIN PRESS sel - 2

PUSH TO TEST PB - PUSH (O2 FLOW INC)

MAIN REG A vlv - open

EMER CABIN PRESS sel - BOTH

5 SEC RAD LEAK CHECK

Monitor SEC ACCUM QUANTITY - no change

(377) SEC GLY To RAD vlv - NORM (CW) for 30 sec,
then BYPASS (CDR)

SYSTEMS CHECKS

ON T027 BOX CUT WITH KNIFE
(AT KNIT NEAREST BULKHEAD)
2 ROPES NEAREST PANEL 351
AND MOVE BOX.

5/24/73 (A.I.)
DATE 5/24/73

6 ECS POST INSERTION CONFIG

(Must be performed between +20:00m & +55:00m)

- (326) GLY RSVR BYPASS vlv - OPEN (CCW)
 GLY RSVR OUT vlv - CLOSE (CW)
 GLY RSVR IN vlv - CLOSE (CW)
 PRIM GLY ACCUM QTY 30-65%
- (379) PRIM ACCUM FILL vlv - ON (CCW) until 40-55%
 ECS RAD FLOW CONT - PWR
 PRIM GLY TO RAD vlv - NORMAL (push)
 ECS RAD HTR - PRIM 1
 ECS RAD TEMP PRIM OUT below PRIM IN
 After 5 min, if outlet temp > inlet:
- * PRIM GLY TO RAD vlv - BYPASS(pull)*
 - * Recheck in 10 min *
- ECS RAD tb - gray
 GLY EVAP TEMP IN - AUTO
 POT H2O HTR - MNA
 cb H2O/URINE DUMP HTR (2) - close

7 SM RCS MONITORING CHECK

- SM RCS PSM 1 He tb - bp
 SM RCS PSM 1 MANF ISOL tb - gray
 SM RCS QUAD He (4) tb - gray
 SM RCS PSM PRPLNT (4) tb - bp
 SM RCS QUAD PRPLNT (8) tb - gray
 RCS IND sel - SM QUAD A, B, C, D, PSM 1
 PKG TEMP - 115-175°F (PSM - zero)
 He TK PRESS - 4100-4200 psia (PSM:3975-4285)
 FUEL MANF PRESS - 178-192 psia (PSM:130-150)
 OXID MANF PRESS - 178-192 psia (PSM:150-170)
 RCS IND sw - TK PRESS/QTY
 FUEL TK PRESS - 178-192 psia (PSM:130-150)

DATE 3/16/73

8 CM RCS MONITORING CHECK

CM RCS PRPLNT tb (2) - gray
RCS IND sw - CM 1,2
He TEMP - 60°-80°F
He PRESS - 4100-4200 psia
MANF PRESS - 80-105 psia

9 C/W OPERATIONAL CHECK

C/W LAMP TEST - 1 (LH MA & 18 lts)
C/W LAMP TEST - 2 (RH MA & 17 lts)
C/W CSM - CM (CM RCS 1t (2) - on)
C/W CSM - CSM (CM RCS 1t (2) - out)
C/W MEMORY - RESET

10 EPS MONITORING CHECK

Cryogenic Pressure - Quantity Check
H2 PRESS (2) - 225-260 psia
O2 PRESS (2) - 865-935 psia
SURGE TK PRESS - 865-935 psia
H2 FANS (2) - AUTO (verify)

FC Power Plant Check

FC HTRS(2) - on(up)
FC REAC tb (2) - gray
H2 VENT tb - gray
FC IND sel - 1, 3

H2 FLOW - 0.03-0.15 lb/hr

O2 FLOW - 0.25-1.2 lb/hr

MOD SKIN TEMP - 390-440° F

MOD COND EXH TEMP - 150-175° F

FC pH HI tb - gray

D-C Voltage-Amperage Check

MN BUS TIE (2) - OFF (verify)

SM PWR SOURCE MNA tb - 1 gray, 2 & 3 bp

SM PWR SOURCE MNB tb - 1 & 2 bp, 3 gray

CSM/SWS INTERFACE PWR tb - gray

SM SOURCE 1 & 3 (check amps)

MAIN BUS A, B, (26.5-31 vdc)

BAT BUS A, B, & BAT C (31.5-38 vdc < 3 amp)

PYRO BAT A, B (36.5 - 37.5 vdc)

DC IND sel - MNB

SYS TEST 3B (BAT RLY BUS - 3.4-4.1 vdc)

A-C VOLTS - 113 to 117 all phases

3/16/73

DATE

11 ECS MONITORING CHECK

SUIT COMP ΔP - .3-.4 psid
02 SURGE TANK PRESS - 865-935 psia
REPRESS 02 >865 psia
PRIM RAD tb - gray

*If PRIM RAD tb - 2: *

* ECS RAD FLOW AUTO CONT - 1 until*

* tb gray, then AUTO

ECS RAD TEMP PRIM IN - 67-97° F
ECS RAD TEMP PRIM OUT - -12° to +65° F
PRIM GLY EVAP TEMP OUT - 38-50.5° F
PRIM GLY DISCH PRESS - 40-52 psig
SUIT TEMP - 45-55° F
SUIT PRESS/CABIN PRESS - 4.7-5.3 psia
PART PRESS CO2 < 7.6 mm Hg
POT H2O QTY - 10-50%
WASTE H2O QTY - 10-50%

12 SECONDARY GLYCOL LOOP CHECK

ECS IND sw - SEC
SEC COOL LOOP PUMP - AC1
GLY DISCH SEC PRESS - 39-51 psig
ACCUM SEC QTY IND - 30-60%
SEC EVAP H2O FLOW - AUTO
SEC COOL LOOP - EVAP

After 5 min:

SEC EVAP TEMP OUT - 38-50.5°F
SEC COOL LOOP EVAP - RSET 1 min, off (ctr)
SEC EVAP H2O FLOW - off (ctr)
SEC COOL LOOP PUMP - off (ctr)
ECS IND sw - PRIM

DATE 3/16/73

L

3-5

SPS TEMP IND sw - He PNL, OXID LN (45-80°F)

13 SPS MONITORING CHECK~~SPS TEMP ind - +50 +3°F (+55 +3°F for SEC)~~

SPS PRESS IND sw - He, N2A, & N2B

SPS PRESS ind - 3750

He 3900 psia max

N2A 2900 psia max

N2B 2900 psia max

SPS PRESS IND sw - He

FUEL & OXID PRESS ind - 170 to 195 psia

SPS ENG INJ VLVS (4) - CLOSE

SPS He VLV (1&2) - AUTO, tb - bp

14 EXTEND DOCKING PROBE

cb DOCK PROBE (2) - close (verify)

DOCK PROBE EXTD/REL - EXTD/REL until

full probe extension

(DOCK PROBE tb - gray at full extension)

EXT RET

FULL EXT	Gray	Gray
FULL RET	BP	BP
PART EXT	BP	Gray

DOCK PROBE EXTD/REL - RETRACT (tb-gray)

3/16/73
5/8/73 (Pfl)

DATE

121

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ABORT PROCEDURES

MODE IA ABORT
(00:00 to 01:01)

00:00 THC - CCW then NEUTRAL
CM/SM SEP (2) - on (up)

00:14 ELS - AUTO
ELS LOGIC - on (up)
TWR JETT (2) - on (up)
APEX COVER JETT PB - PUSH

00:16 *DROGUE DEPLOY PB - PUSH*

00:18 *CM RCS He DUMP PB - PUSH*

Monitor altimeter

If <alidade - DEPLOY MAINS
>alidade - NO ACTION

00:28 If <10,000 ft - DEPLOY MAINS

Note: Alidade set for 3800 ft true altitude
prior to Launch

Go to LANDING PHASE pg L/4-11

MODE IB ABORT
(01:01 to 16.5 nm)

00:00 THC - CCW then NEUTRAL
CM/SM SEP (2)-on (up)

00:11 *CANARD DEPLOY PB - PUSH*

00:14 ELS - AUTO
ELS LOGIC - on (up)
RCS CMD - ON

Go to LANDING PHASE pg L/4-11

MODE IC ABORT

(16.5 nm to TWR JETT)

00:00 THC - CCW then NEUTRAL
CM/SM SEP (2) - on (up)
RCS CMD - ON

00:11 *CANARD DEPLOY PB - PUSH*
CM RCS PRESS - on (up)
RCS TRNFR - CM
RCS IND - CM (1 or 2)
C/W MODE - CM

S/C PLATFORM GO/NO GO
KEY RLSE to N44, Check HA

HA>32nm & PLAT GO

HA<32nm or PLAT NO GO

TWR JETT sw(2)-on(up)
MAN PITCH - RATE CMD
BMAG (3)- ATT1/RATE 2
ENT ATT R0°,PI35°,Y0°
EMS FUNC - ENTRY
EMS MODE - NORMAL
At .05G Lt,
.05G sw - on (up)
Fly Max Lift

Cmd +5°/sec Pitch rate

If +Pitch rate too hi:

* Roll 90° *
* Damp rate with yaw *
* Roll to HDS DN *

P (.05G) _____
GET DRO _____

Go to LANDING PHASE pg L/4-11

LET FAILS TO JETTISON

LEGS CUT/NO MOTOR FIRE (pyro audible):

LES MOTOR FIRE PB - PUSH

NO RESPONSE to TWR JETT switches:

cb SECS ARM (2) - close (verify)

cb SECS LOGIC (2) - close (verify)

cb EDS (3) - close (verify)

SECS LOGIC (2) - on (up) (verify)

SECS PYRO ARM (2) - on (up) (verify)

EDS PWR - on (up) (verify)

TWR JETT (2) - on (up)

NO TWR JETT: TWR JETT (2) - AUTO

Abort, Mode IC on STDN cue

DATE 2/5/73

MODE II RCS ABORT
(TWR JETT to MODE III)

00:00 THC - CCW (4 sec min)
If No BECO: - Reset THC
* Req. RSO shutdown *
* Reset & start DET *

00:03 *CSM/LV SEP - PUSH*
*RCS CMD - ON *

THC - ARMED
00:05 THC - NEUTRAL, Damp rates, then +X
Check SM RCS talkbacks

00:24 THC - +X OFF
BMAG MODE (3) - ATT1/RATE 2
KEY RLSE to N44, Check TFF
If TFF > 2 min, Yaw 45° (LEFT) out-of-plane
cb MNA&B BAT C (2) - close
CM/SM SEP - on (up)
CM RCS PRESS - on (up)
RCS TRNFR - CM
C&W MODE - CM
Entry ATT - (R=0°, P=130°, Y=0°) (Compl by 1:40)
cb DOCK RING SEP (2) - close (pull lanyard)
DOCK RING SEP (2) - on (up)
EMS FUNC - ENTRY GET 300K _____
EMS MODE - NORMAL P (.05G) _____
GET DRO _____
At .05G lt - on
.05G sw - on (up)
EMS ROLL - on (up)
Fly Max Lift
N62E VI, HDOT, H

Go to LANDING PHASE pg L/4-11

DATE 3/16/73

MODE II, MODE IIIA

MODE IIIA SPS ABORT (Burn STDN Pad)
($\Delta R = -1150\text{nm}$ to $\Delta R = -450\text{nm}$)

00:00 THC - CCW (4 Sec Min)
If No BECO: - Reset THC
* Req. RSO Shutdown *
* Reset & start DET *

00:03 *CSM/LV SEP - PUSH*
*RCS CMD - ON *

THC - ARMED
00:05 THC - NEUTRAL, Damp rates, then +X
Check SM RCS talkbacks

00:24 THC - +X OFF
N5OE ΔR , HP, TFF (.1nm, min-sec)
BMAG MODE (3) - ATT1/RATE2
MNVR to Burn att (180° , 355° , 0°)
(Scribe on horiz, SEF, Hds dn)
EMS MODE - NORMAL
RATE - LOW
LV/SPS IND - GPI

ΔV THRUST A - NORMAL
01:50 THC - +X (15 sec)
02:05 THRUST ON PB - PUSH
Burn to VC ($\Delta R = -450$)
or TFF = 1+40
 ΔV THRUST (2) - OFF

TIG
(1999.9)
 ΔV
CUTOFF
BT
P(IGN)
GET 300K
P (.05G)
GET DRO

3/16/73
DATE

RATE - HIGH
If TFF > 2min, Yaw 45° (LEFT)
out-of-plane
cb MNA&B BAT C(2) - close
CM/SM SEP - on (up)
CM RCS PRESS - on (up)
RCS TRNFR - CM
C&W MODE - CM
Mnvr to entry att ($R=0^\circ$, $P=115^\circ$, $Y=0^\circ$)
(BEF, Hds Dn, Full Lift)
cb DOCK RING SEP (2) - close (pull lanyard)
DOCK RING SEP (2) - on (up)

MODE II, MODE IIIA

Note TFF

EMS MODE - STBY

EMS FUNC - ENTRY

EMS MODE - NORMAL

At .05G lt - on

.05G sw - on (up)

EMS ROLL - on (up)

Fly Max Lift

Go to LANDING PHASE, pg L/4-11

DATE 2/5/73

MODE IIIB

MODE IIIB SPS ABORT (Burn STDN Pad)
($\Delta R = -450$ nm to INSERTION)

00:00 THC - CCW (4 Sec Min)
If No BECO: - Reset THC
* Req. RSO shutdown *
* Reset & start DET *

00:03 *CSM/LV SEP - PUSH*
*RCS CMD - ON *

THC - ARMED
00:05 THC - NEUTRAL, Damp rates, then +X
Check SM RCS talkbacks

00:24 THC - +X OFF
N50E ΔR , HP, TFF (.1nm,min-sec)
BMAG MODE (3) - ATT1/RATE2
If $\Delta R > -450$:
MNVR to retro att ($R=180^\circ, P=198^\circ, Y=0^\circ$)
(Scribe on horiz, BEF, Hds up)
EMS MODE - NORMAL
RATE - LOW
LV/SPS IND - GPI

ΔV THRUST A - NORMAL
01:50 THC - +X (15 sec)
02:05 THRUST ON PB - PUSH
Burn to VC ($\Delta R = -450$)
or TFF = 1+00
 ΔV THRUST (2) - OFF

TIG _____
(1999.9) _____
 ΔV _____
CUTOFF _____
BT _____
P (IGN) _____
GET 300K _____
P (.05G) _____
GET DRO _____

RATE - HIGH
If TFF > 2min, Yaw 45° (LEFT)
out-of-plane

cb MNA&B BAT C(2) - close
CM/SM SEP - on (up)
CM RCS PRESS - on (up)
RCS TRNFR - CM
C&W MODE - CM

MNVR to entry att ($R=0^\circ, P=115^\circ, Y=0^\circ$)
(BEF, Hds Dn, Full Lift)

cb DOCK RING SEP (2) - close (pull lanyard)
DOCK RING SEP (2) - on (up)

MODE IIIB

DATE 2/5/73

Note TFF

EMS MODE - STBY

EMS FUNC - ENTRY

EMS MODE - NORMAL

At .05G lt - on

.05G sw - on (up)

EMS ROLL - on (up)

Fly Max Lift

*If TFF C/O before $\Delta R = -450$: *

* At .2G, Roll right 90° (LV north)*

Go to LANDING PHASE pg L/4-11

DATE 4/6/73

MODE IV SPS TO ORBIT

(VI ~ 24,470, HDOT ~ -145, H ~ 84)

00:00 THC - CCW (4 sec min)
 *If No BECO: - Reset THC *
 * Req. RSO shutdown *
 * Reset & start DET *

00:03 *CSM/LV SEP - PUSH*
 *RCS CMD - ON *

THC - ARMED
 00:05 THC - NEUTRAL, Damp rates, then +X
 Check SM RCS talkbacks

00:24 THC - +X OFF

Perform PITCH PROFILE or FIXED ATTITUDE BURN:

PITCH PROFILE

BMAG MODE (3) - ATTI/RATE2
 MnvR to Chart Burn Attitude
 EMS MODE - NORMAL
 RATE - LOW
 LV/SPS IND - GPI
 ΔV THRUST A - NORMAL
 THC - +X (15 sec)
 <01:30 THRUST ON PB - PUSH
 SCS TVC (PITCH) - RATE CMD
 Fly HDOT to zero (+100 fps), trim YAW with tw
 Burn to chart VI, pg L/2-6

*If unable to fly HDOT to <|100|: *
 * KEY RLSE to N44 *
 * (-HDOT) Burn HP >70nm +4 sec *
 * (+HDOT) Burn HA >200nm, then Apogee Kick*

ΔV THRUST (2) - OFF
 EMS MODE - STBY

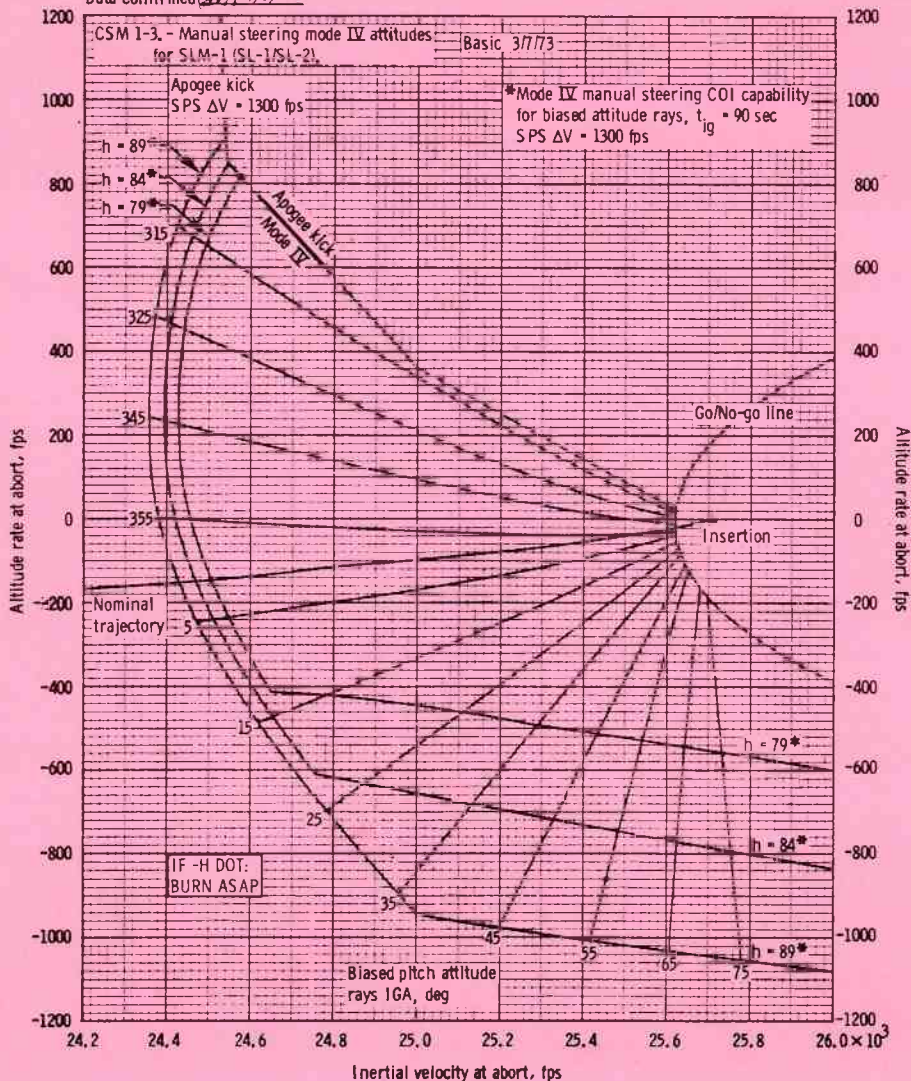
Go to "Log VI", pg L/4-10

Henderson/FPB/MPAD

SL-2(SA-206) LV Operational Traj. (MSC Tape #A04529)

Data source SL 3DB Vol. II Rev B Oct. 1972 & Vol. III Amend. #106 Nov. 3, 1972

Data confirmed 2/11/5/7



or FIXED ATTITUDE BURN (Scribe on horiz, SEF, Hds Dn)

BMAG MODE (3) - ATT1/RATE2

EMS MODE - NORMAL

RATE - LOW

LV SPS IND - GPI

 ΔV THRUST A - NORMAL

01:50 THC - +X (15 sec)

02:05 THRUST ON PB - PUSH

BURN to VC

 ΔV THRUST (2) - OFF

EMS MODE - STBY

TIG

1999.9

 ΔV

CUTOFF

BT

P(IGN)

Log VI _____ (fps)

H DOT _____ (fps)

H PAD _____ (.1nm)

KEY RLSE

Log HA _____ (.1nm)

HP _____ (.1nm)

TFF _____ (min-sec)

PRO

V37E 00E

Load DAP, V48: R1=11102, R2=01111

V46E

GMBL MTRS (4) - OFF (PLT confirm)

CM RCS LOGIC - OFF

cb ELS/CM-SM SEP (2) - open

cb FLT/PL VENT - open

MN BUS TIE (2) - OFF

CAB PRESS REL vlv (2) - NORMAL/LATCHED

PCM BIT RATE - LOW

BMAG MODE (3) - RATE 2

SC CONT - CMC/AUTO

Go to POST SEPARATION, pg L/2-11

DATE 2/5/73

LANDING PHASE (30K, DESCENDING)

30K' ELS LOGIC - on (up)
ELS - AUTO

24K' Twr jett (auto)
*TWR JETT (2) - on (up) *
Apex cover jett (auto)
APEX COVER JETT PB - PUSH)
(wait 2 sec)

Drogues deployed (auto)
DROGUE DPLY PB - PUSH

*If Both Drogues Fail: *
* ELS - MAN *
* STABILIZE CM (DIRECT RCS)*
* 5K' MAIN DPLY PB - PUSH *
* ELS - AUTO *

46 sec

23.5K' Cabin Pressure increasing
*If not increasing by 17K': *
* CABIN PRESS REL viv (RH) - DUMP*

CM RCS PRPLNT (2) - OFF

10K' Main parachutes deployed (Cab Press = 10 psia)
MAIN DEPLOY PB - PUSH
VHF ANT - RECY
VHF AM A - SIMPLEX
VHF BCN - ON

*If No Comm and abort occurred between *
* 1:01 & 2:00 min or if land impact ex-*
* pected: *
* Perform CM RCS DUMP, pg L/4-12 *

LANDING PHASE

DATE 2/5/73

- CABIN PRESS REL vlv (RH) - DUMP
 STRUT LOCKS (4) - UNLOCK
 (5) cb FLT & PL BAT BUS A,B,&BAT C (3) - close
 cb FLT & PL MNA & B (2) - open
 cb BAT RLY BUS (2) - open
 (8) cb SPS P&Y (4) - open

ELS - AUTO (verify)
 ELS LOGIC - on (up) (verify)
 FLOOD Lts - POST LDG

- 800' CAB PRESS REL vlv (2) - CLOSE (latch off)
 DIRECT O2 vlv - OPEN (CCW)
 MN BUS TIE (2) - OFF

Go to POST LANDING PROCEDURES, pg L/4-13

CM RCS DUMP; if req'd (land landing)

- CABIN PRESS REL vlv (2) - CLOSE
 CM RCS LOGIC - on (up)
 If main or pyro bus lost:
 * Use RHC's for burn, *
 * not DUMP sw *
 CM PRPLNT - DUMP (burn audible)
 MONITOR CM RCS 1&2 for He press decrease
 If no burn or press decrease:
 * Use both RHC's *
 * DO NOT FIRE PITCH JETS *
 CM PRPLNT - PURGE
 *CM RCS He DUMP PB - PUSH *
 RHC (2) - 30 secs, NO PITCH

DATE 2/5/73

POST LANDING

POST LANDING COMMUNICATIONS

VHF ANT - RECY (verify)

VHF BCN - ON (verify)

If no contact with recovery forces:

* Perform VHF BEACON Check *

MONITOR VHF BEACON transmission with

VHF AM B Rcvr and/or Survival Trncvr (VOICE)

*If VHF Beacon not operating:

- * Connect Survival Trncvr cable conn *
- * J1 to bcn ant cable conn P112 behind VHF*
- * ant access pnl and place radio in BCN *
- * mode *

LOW POWER CHECKLIST

VHF BCN - OFF

VHF AM (3) - RCV

FLOOD LTS - OFF

VHF AM A&B - off (ctr)

VHF AM RCV ONLY - A (verify)

POSTLANDING VENT SYS: minimize use

SURV RADIO - plug into VHF BCN ANT cable

conn P112 behind VHF ant access pnl & turn
radio on in BCN mode

*If BAT C <27.5 vdc: *

- * cb BAT BUS A BAT A - open *
- * cb BAT BUS A PYRO BAT A - close*
- * cb FLT/PL BUS BAT A - close *
- * cb FLT/PL BUS BAT C - open *
- * Monitor PYRO BAT A voltage *
- * on BAT BUS A *

*If PYRO BAT A <27.5 vdc: *

- * cb BAT BUS B BAT B - open *
- * cb BAT BUS B PYRO BAT B - close*
- * cb FLT/PL BUS BAT B - close *
- * cb FLT/PL BUS BAT A - open *
- * Monitor PYRO BAT B voltage *
- * on BAT BUS B *

NOMINAL EGRESS & POWER DOWN

- (15) PL VENT - OFF
- (5) cb FLT/PL BUS BAT C (1) - open
- cb BAT CHRG BAT C/EDS 2 (1) - open
- cb Pn1 275 (all) - open
- Charge hatch counterbalance
- Open side hatch (after collar installed)
- ACTR HNDL SEL - N
- GN2 vlv HNDL - VENT (pull)
- GN2 vlv HNDL - PRESS (push)
- Check Pressure Gauge (mid-white)
- *Repeat vent/press to obtain mid-white*

DATE 2/5/73

UNAIDED EGRESS PROCEDURES

- * If no ventilation or CM O2 supply: *
- * If Stable I, open side hatch as req'd *
- * If Stable II, initiate egress within 2-1/2 hrs

PREPARATION

- Disconnect umbilicals
- (F2) Neck dams on (if suited)
- Configure couch(es) - 270°
- Armrests stowed
- (R4) Unstow rucksacks #1 & #2
- Connect lanyards
- (yellow to rucksack, green to S/C, white to crew, in order of egress printed on lanyards)

STABLE I

- (15) PL VENT - OFF
- (5) cb FLT/PL BUS BAT C (1) - open (verify)
- cb BAT CHRG BAT C/EDS 2 (1) - open
- cb PANEL 275 (all) - open
- Charge hatch counterbalance
- Open side hatch
- ACTR HNDL SEL - N
- GN2 vlv HNDL - VENT (pull)
- GN2 vlv HNDL - PRESS (push)
- Check pressure gauge (mid-white)
- *Repeat vent/press to obtain mid-white*
- Remove raft from rucksack #2
- Put raft overboard & pull inflation lanyard
- Pass hardware kit to raft
- Egress, inflate life vest, board raft

STABLE II

- (6,9,10) PWR (3) - OFF
- SUIT PWR (3) - OFF
- Review hatch handle location and hatch unlocking procedure
- PRESS EQUAL vlv - OPEN
- Remove and stow tunnel hatch on underside of CDR's couch
- Lower hardware rucksack down tunnel
- Exit feet first; when clear of S/C inflate water wings
- Remove life raft from rucksack #2 and inflate

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EMERGENCY LV SEPARATION (INSERTION TO SIVB SEP)

THC CCW (4 SEC)
THC - ARMED
THC - NEUTRAL, then +X & hold thru Ign
MNVR TO BURN ATT (if reqd) (not retrograde)
MN BUS TIE (2) - ON
BMAG MODE (3) - ATT1/RATE 2
GMBL MTRS (4) - ON
 ΔV THRUST A - NORMAL
THRUST ON PB - PUSH
SPS BURN (3 SEC) - THEN ΔV THRUST A - OFF

DATE 4/3/73

EMERGENCY OA RATE CONTROL

IF ATM UNABLE TO CONTROL RATES

MODE EXP PTG - STBY, tb - STY
MOMENTUM DUMP CAGE - ZERO H, tb - CGE

CDR transfer to CSM
cb SCS LOGIC BUS (4) - close
BMAG MODE (3) - RATE 1
LOGIC 2/3 PWR - on (up)
ELEC PWR - GDC/ECA
SIG CONDR/DR BIAS PWR (2) - AC1/AC2
BMAG PWR (2) - ON
FDAI GPI PWR - 1
RHC PWR NORM (2) - AC/DC
RHC PWR DIR 2 - MNA/MNB
FDAI - 1/GDC
MAN ATT (3) - RATE CMD
THC PWR - on (up)
SC CONT - SCS
SM RCS PSM 1 He - OPEN, tb - gray
SM RCS QUAD He (4) - CLOSE, tb (4) - bp
SM RCS QUAD PRPLNT (4) - CLOSE, tb (8) - bp
SM RCS PSM PRPLNT (4) - OPEN, tb (4) - gray
cb SCS A/C ROLL, B/D ROLL, PITCH, YAW MNA(4)-close
AUTO RCS (16) - MNA
RHC, THC - ARMED

Stop OA P&Y rates w/THC (fly FROM rate needles)
Stop OA Roll rates w/RHC (fly TO needle)

Monitor rates decreasing at ATM C&D pnl

If OA venting
Attempt to isolate and correct

IF RATES CONTINUE TO DIVERGE

SPT & PLT transfer to CSM
Perform EMERGENCY OA UNDOCKING

DATE 5/19/73

EMERGENCY OA UNDOCKING

If Fuel Cells Shutdown

SM PWR SOURCE (2) - BAT 1 & 3
SM PWR SOURCE 1 MNA - on(up), tb - gray
SM PWR SOURCE 3 MNB - on(up), tb - gray
SM PWR DISC 1t - mom
CSM/SWS INTERFACE PWR (2) - DISC, tb - gray
cb BAT BUS A&B BAT A&B (2) - close (Pnl 275)
cb BAT BUS A&B MAIN A&B (2) - open (Pnl 275)

cb SCS LOGIC BUS (4) - close
BMAG MODE (3) - RATE 1
LOGIC 2/3 PWR - on(up)
ELEC PWR - GDC/ECA
SIG CONDR/DR BIAS PWR (2) - AC1/AC2
BMAG PWR (2) - ON
FDAI/GPI PWR - 1
RHC PWR NORM (2) - AC/DC
RHC PWR DIR 2 - MNA/MNB
FDAI - 1/GDC
MAN ATT (3) - RATE CMD
THC PWR - on(up)
SC CONT - SCS
SM RCS PSM 1 - He - open, tb - gray
SM RCS QUAD He (4) CLOSE, tb(4)-bp
SM RCS QUAD PRPLNT (4) - CLOSE, tb (8) - bp
SM RCS PSM PRPLNT (4) - OPEN, tb (4) - gray
cb SCS A/C ROLL, B/D ROLL, PITCH, MAW MNA (4) - close
AUTO RCS (16) - MNA

Remove Air Interchange Duct & toss in MDA
Close MDA Hatch
Verify MDA PRESS EQUAL vlv - CLOSE
Disconnect Drag Cable & place in MDA tunnel
CABIN PRESS RELF vlv (2) - NORM
CABIN REPRESS vlv - OFF (CCW)
MAIN REG vlv (2) - OPEN
EMER CAB PRESS vlv - BOTH
SURGE TK 02 vlv - ON
Install Tunnel Hatch (decal)
Hatch Integrity Check (decal)(If time permits)

EMER OA UNDOCK

DATE 5/19/73

RHC, THC - ARMED
 cb SECS ARM (2) - close
 cb DOCK RING SEP (2) - close (Pnl 278)
 SECS LOGIC (2) - on(up)
 SECS PYRO ARM (2) - on(up)
 0:00 THC - -X (4 jet)
 0:03 DOCK RING SEP (2) - on(up)
 0:27 THC - neutral
 BMAG MODE (3) - ATT 1/RATE 2
 Station keep at safe distance
 SECS PYRO ARM (2) - SAFE
 SECS LOGIC (2) - OFF
 cb SECS ARM (2) - open
 SUIT CRT RET vlv - open (pull)
 Remove CDR hose interconnect
 DEMAND REG sel - BOTH
 Install LiOH canisters (A4,A6, or TSB)
 SUIT COMPR 1 - ACT

POST EMER OA UNDOCKING

Unstow SWS DEACT C/L (R3) & perform:	Page
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POST EMER
OA UNDOCK

DATE 4/3/73

ECS

SUIT COMPRESSOR LITE - CLOSED SUIT LOOP

SWITCH TO OTHER COMPRESSOR ON OTHER BUS
SEE ECS 9

O2 FLOW HI + RAPID LOSS OF SURGE TK PRESS
+ CABIN PRESS <4.6 PSI

CABIN PRESS RELF vlv (2) - CLOSE
✓TUNNEL EQUALIZATION vlv - CLOSED
REPRESS PKG vlv - ON (WHEN SURGE TK PRESS <150 PSI)
✓EMERG CABIN PRESS REGS - BOTH
DON SUITS

CONTAMINATION IN CM

DON O2 MASKS

CONTAMINATION IN CLOSED SUIT LOOP

CHANGE TO OTHER SUIT COMPR
DIRECT O2 vlv - FULL OPEN THEN ADJUST FOR SUIT
TO CABIN ΔP OF 2 IN OF H2O

IF CONDITION PERSISTS

SUIT COMPR (2) - OFF
DOFF HELMETS
DIRECT O2 vlv - CLOSE
DON O2 MASKS

FIRE/SMOKE IN CM (DOCKED)

IF CM CAN BE ENTERED

MDA FANS CSM - OFF (AM 203)
ATM/TRANSFER TIE BUS 1&2 (2) - OPEN (mom)
REG/TRANSFER TIE BUS 1&2 (2) - OPEN (mom)
ALL CREWMEN DON SOMAS
ONE CREWMAN ENTER CM
SM O2 SUPPLY - OFF
IVA STA O2 SUPPLY - OFF (verify)
CAB FANS (2) - OFF (verify)
SM PWR SOURCE MNA & MNB (6) - OFF
cb BAT BUS A BAT A - open (verify)
cb BAT BUS B BAT B - open (verify)
USE FIRE EXTINGUISHER

DATE 2/12/73

ECS, FIRE/SMOKE,
CRIT BURNS (OVER)

IF FIRE PERSISTS

CAB PRESS RELF vlv - DUMP
EXIT CM WITH AID
CLOSE MDA HATCH
DOFF SOMAS ONLY AFTER SMOKE REMOVED
IF CM CANNOT BE ENTERED

MDA FANS CSM - OFF (AM 203)
ATM/TRANSFER TIE BUS 1&2 (2) - OPEN (mom)
REG/TRANSFER TIE BUS 1&2 (2) - OPEN (mom)
REMOVE AID
CLOSE MDA HATCH

FIRE/SMOKE IN CM (UNDOCKED)

MONITOR DC FOR HI CURRENT - REMOVE POWER
FROM ASSOCIATED INVERTER
IF CURRENT REMAINS HI - REMOVE POWER FROM
ASSOCIATED DC BUS
IF CLOSED SUIT LOOP, SWITCH SUIT COMPR TO GOOD AC BUS
IF HELMET OFF, SUIT COMPR (2) - OFF
RECONFIGURE INVERTER 3 ON LOST AC BUS
VERIFY RCS CONTROL POWER CONFIGURATION
IF HELMETS OF
DON O2 MASKS
USE FIRE EXTINGUISHER OR H2O GUN (OPTIONAL)
IF CLOSED SUIT LOOP
USE FIRE EXTINGUISHER OR H2O GUN (OPTIONAL)
✓ EMERG CABIN PRESS REGS - OFF
IF FIRE PERSISTS - DUMP CABIN

G&N CRITICAL BURNS

IF NO START OR ISS LITE + PROG LITE
IF CMC LITE, PROG ALARM 1407 OR EARLY CUTOFF

SCS TVC (2) - AUTO
SC CONT - SCS
✓ ATTITUDE
SPS THRUST - DIRECT (MOMENTARY), IF REQ'd

IF ABNORMAL DYNAMICS

THC CW, control rates by MTVC
After SHUTDOWN, AUTO RCS (16) - OFF

SCS CRITICAL BURN

IF NO START OR EARLY CUTOFF

SPS THRUST - DIRECT (MOMENTARY)

IF RATE NEEDLE HARDOVER & FDAIs DIVERGE OPPOSITE

BMAG MODE (3) - RATE 1

THC - CW, use MTVC

IF ABNORMAL DYNAMICS IN AUTO MODE

THC - CW, use MTVC

BMAG MODE (3) - RATE 2

IF ABNORMAL DYNAMICS IN MTVC MODE

THC - CW

IF PROBLEM PERSISTS, SHUTDOWN

AUTO RCS (16) - OFF

SPS

IF NO CUTOFF AFTER ΔV THRUST (BOTH) - OFF

cb SPS PILOT VLVS - open

IF EMS & N40 (R3) STILL COUNTING AFTER SHUTDOWN

SC CONT - SCS

TRANS CONT PWR - OFF

cb DIR ULLAGE (2) - open

IF CONDITION PERSISTS, AUTO RCS SEL (16) - OFF

SM RCS PRPLNT (AFFECTED QUAD) - OFF

SPS PRESS LITE

CONTINUE CRITICAL BURN

IF FUEL & OX PRESS (BOTH) > 200 PSI

SPS HE vlvs (2) - OFF, THEN CONTROL MANUALLY
BETWEEN 170-200 PSI

IF FUEL/OX ΔP > 20 PSI

SPS HE vlvs (2) - ON

IF CONDITION PERSISTS, SPS HE vlvs(2)-OFF(Until
Pc < 70)

G&C (COASTING, ENTRY)

CMC LITE

SC CONT - SCS, SEE G&N 5

ISS LITE + PROG ALARM LITE

SC CONT - SCS, SEE G&N 6

DATE 2/12/73

G&C, SPS,
EPS (OVER)

EMERGENCY POWER DOWN

INSURE DSE IS RECORDING

DC AMPS

IF UNSUITED, SUIT COMP (2) - OFF

4.0

FC PUMPS (2) - OFF (Until Tskin > 475°F)

5.8 TOTAL

cb G&N OPTICS MNA & MNB (2) - OPEN (Pn1 5)

3.1

G&N PWR (AC) - OFF (Pn1 100)

0.9

O2 HTRS (2) - OFF (CTR)

11.4

H2 HTRS (2) - OFF (CTR)

1.4 TOTAL

H2 FANS (2) - OFF (CTR)

1.0

C/W NORMAL - ACK

ECS RAD HTRS (2) - OFF

17.2 EA

POT H2O HTR - OFF

1.6 MAX

SM RCS QUAD HTRS (4) - OFF

21.4 MAX

SM RCS ENGINE HTRS (4) - OFF

3.3 MAX EA

PSM 1 HTR - OFF

1.4 MAX

LIGHTS - Min Req'd

5.3 MAX

EXT LTS - OFF

4.6

VHF RANGING - OFF

1.4

S BD AUX TV - OFF (CTR)

5.3

SPS HTRS - OFF (CTR)

6.5 MAX

SIG CONDR/DRIVER BIAS PWR (2) - OFF

SECURE ONE BMAG

2.6

SELECT SINGLE JET CONTROL

EMS FUNC - OFF

RHC PWR DIRECT (2) - OFF

THC PWR - OFF

CONFIGURE FOR SINGLE INVERTER OPERATION

TURN OTHER INVERTER OFF

4.0 MAX

BAT CHGR - OFF

NOTE MISSION TIME

cb TIMERS (2) - OPEN (Pn1 5)

AC INVERTER (9) - OFF

CM RCS HTRS - OFF

ISOLATE FAILED FC's from MAIN BUSES

4/3/73
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EMER
1-9

ECS POWER DOWN	3.7 TOTAL
ECS GLY PUMP sel - OFF (ISS LIMIT 2.5 HRS)	2.6
ECS RAD FLOW CONT PWR - off (CTR)	0.7
GLY EVAP TEMP IN - MAN	
ECS RAD HTRS (2) - OFF	
GLYCOL EVAP H2O FLOW - OFF	~0.1
GLYCOL EVAP STEAM PRESS - MAN	~0.2

COMM POWER DOWN	13.0 TOTAL
IF VOICE DESIRED	
UP TLM CMD RESET - RESET then OFF	
S-BD AUX TAPE - DN VOICE BU	
S-BD MODE PCM - OFF	
PCM BIT RATE - HIGH	
S-BD PWR AMP - OFF (CTR)	4.0
TAPE RCDR - OFF (CTR)	1.6
SCE PWR - OFF (CTR)	0.7
cb INSTR MNA & MNB (2) - OPEN (Pnl 5)	4.9
TELCOM GRP 1 & 2 (2) - OFF	1.6

CMC/IMU POWER DOWN	6.0 IMU
COMPLETE ALIGNMENT TRANSFER	
CMC MODE - FREE	PROVIDES CMC MIN IMP
cb G&N IMU MNA & MNB (2) - OPEN (Pnl 5)	
V37E06E	3.0 CMC
F V50 N25, 00062, CMC PWR DN	
PRO, HOLD (~5 SEC) UNTIL STBY LT - ON	

SCS POWER DOWN	6.0
ACCEPTABLE S/C ATTITUDE	
BMAG PWR (2) - OFF	
FDAI/GPI PWR - OFF	PROVIDES MIN IMP
SCS ELECTRONICS PWR - ECA	(REQUIRES ACT & MNB)
ORDEAL PWR & LIGHTING - OFF	
cb SCS LOGIC BUS (4) - OPEN (Pnl 8)	2.0
SCS ELECTRONICS PWR - OFF	
RHC PWR NORM (2) - OFF	

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LAUNCH BUS LOSS

MN BUS A LOST - LAUNCH

EDS AUTO/OFF - OFF
TVC GMBL DR (P,Y) - 2
SCS TVC (P,Y) - RATE CMD
BMAG MODE (3) - RATE 2
FDAI SEL - 2
cb SPS PITCH 2 & YAW 2 (Pnl 8) - OPEN
(AFTER GIMBAL MOTORS ON)

AC INV 3 - MNB
AC INV 3 AC 1 - ON
AC INV 1 AC 1 - OFF
All SM PWR SOURCE MNA - OFF
All SM PWR SOURCE MNB - MNB (BEFORE CM/SM SEP)
cb MNA BAT BUS A (Pnl 275) - OPEN
cb MNB BAT C (Pnl 275) - CLOSED

MN BUS B LOST - LAUNCH

EDS AUTO/OFF - OFF
TVC GMBL DR (P,Y) - 1
SPS TVC (P,Y) - RATE CMD
✓BMAG MODE (3) - RATE 1
FDAI SEL - 1
cb SPS PITCH 1 & YAW 1 (Pnl 8) - OPEN
(AFTER GIMBAL MOTORS ON)

AC INV 3 - MNA
AC INV 3 AC 2 - ON
AC INV 2 AC 2 - OFF
All SM PWR SOURCE MNB - OFF
All SM PWR SOURCE MNA - MNA (BEFORE CM/SM SEP)
cb MNB BAT BUS B (Pnl 275) - OPEN
cb MNA BAT C (Pnl 275) - CLOSED

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AC BUS 1 LOST - LAUNCH

BMAG MODE (3) - RATE 2
FDAI SEL - 2
TVC SERVO PWR 1 - AC2/MNB
SCS TVC PITCH, YAW - RATE CMD

AC INV 1 MNA - OFF
SUIT COMPR - AC 2
ECS GLY PUMP - AC 2
S BD NORM XPNDR - SEC
S BD NORM PWR AMP - SEC

AC BUS 2 LOST - LAUNCH

✓BMAG MODE (3) - RATE 1
FDAI SEL - 1
TVC SERVO PWR 2 - AC1/MNA
MTVC WITH THUMBWHEELS (MODE III OR IV)

AC INV 2 MNB - OFF
✓SUIT COMPR - AC 1
✓ECS GLY PUMP - AC 1

BAT BUS A LOST - LAUNCH

EDS AUTO/OFF - OFF
AUTO RCS SEL (RING 1) - OFF (BEFORE CM/SM SEP)
IF BUS LOST BEFORE GMBL MTRS ON
TVC GMBL DR (P,Y) - 2
cb SPS P2 & Y2 (Pn1 8) - OPEN
(AFTER SEC GIMBAL MOTORS ON)
cb MNA BAT C (Pn1 275) - CLOSED

BAT BUS B LOST - LAUNCH

EDS AUTO/OFF - OFF
AUTO RCS SEL (RING 2) - OFF (BEFORE CM/SM SEP)
IF BUS LOST BEFORE GMBL MTRS ON
TVC GMBL DR (P,Y) - 1
cb SPS P1 & Y1 (Pn1 8) - OPEN
(AFTER PRI GIMBAL MOTORS ON)
cb MNB BAT C (Pn1 275) - CLOSED

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5/8/73 (PFI)

SPS BURN BUS LOSS

MN BUS A LOST - SPS BURN

TVC GMBL DR (P,Y) - 2
SCS TVC (P,Y) - RATE CMD
cb SPS P2 & Y2 (Pnl 8) - OPEN
(AFTER GMBL MTRS ON)
FDAI SEL - 2
✓FDAI SOURCE - CMC
RHC PWR DIRECT 2 - MNB
BMAG MODE (3) - RATE 2
✓ Δ V THRUST B - NORM
AUTO RCS SEL - MNB

AC INV 3 - MNB
AC INV 3 AC 1 - ON
AC INV 1 AC 1 - OFF
All SM PWR SOURCE MNA - OFF
All SM PWR SOURCE MNB - MNB
cb MNA BAT BUS A (Pnl 275) - OPEN

MN BUS B LOST - SPS BURNS

SCS TVC (P,Y) - RATE CMD
TVC GMBL DR (P,Y) - 1
cb SPS P1 & Y1 (Pnl 8) - OPEN
(AFTER GMBL MTRS ON)
FDAI SEL - 1
✓FDAI SOURCE - CMC
RHC PWR DIRECT 1 - MNA
BMAG MODE (3) - RATE 1
✓ Δ V THRUST A - NORM
AUTO RCS SEL - MNA

AC INV 3 - MNA
AC INV 3 AC 2 - ON
AC INV 2 AC 2 - OFF
All SM PWR SOURCE MNB - OFF
All SM PWR SOURCE MNA - MNA
cb MNB BAT BUS B (Pnl 275) - OPEN

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AC BUS 1 LOST - SPS BURNS

TVC SERVO PWR 1 - AC2/MNB
SCS TVC (P&Y) - RATE CMD
BMAG MODE (3) - RATE 2
FDAI SEL - 2
✓FDAI SOURCE - CMC

AC INV 1 MNA - OFF
SUIT COMPR - AC 2
ECS GLY PUMP - AC 2
S BD NORM XPNDR - SEC
S BD NORM PWR AMP - SEC
SPS GAUGING - AC 2

AC BUS 2 LOST - SPS BURNS

TVC SERVO PWR 2 - AC1/MNA
BMAG MODE (3) - RATE 1
SCS TVC (P&Y) - AUTO
MTVC WITH TRIM THUMBWHEELS (SCS BURN ONLY)
FDAI SEL - 1
✓FDAI SOURCE - CMC

AC INV 2 MNB - OFF
✓SUIT COMPR - AC 1
✓ECS GLY PUMP - AC 1

BAT BUS A LOST - SPS BURNS

TVC GMBL DR (P,Y) - 2
(IF BUS LOST BEFORE GMBL MTRS ON)
cb SPS P2 & Y2 (Pn1 8) - OPEN
(AFTER GMBL MTRS ON)
cb MNA BAT C (Pn1 275) - CLOSED

BAT BUS B LOST - SPS BURNS

TVC GMBL DR (P,Y) - 1
(IF BUS LOST BEFORE GMBL MTRS ON)
cb SPS P1 & Y1 (Pn1 8) - OPEN
(AFTER GMBL MTRS ON)
cb MNB BAT C (Pn1 275) - CLOSED

DATE 2/12/73

ENTRY BUS LOSS

MN BUS A LOST - ENTRY

BMAG MODE (3) - RATE 2
FDAI SEL - 2
✓FDAI SOURCE - CMC
AUTO RCS SEL (12) - MNB (ONLY IF BUS LOST AFTER SM SEP)
AC INV 3 - MNB
AC INV 3 AC 1 - ON
AC INV 1 AC 1 - OFF
A11 SM PWR SOURCE MNA - OFF (BEFORE CM/SM SEP)
ALL SM PWR SOURCE MNB - MNB (BEFORE CM/SM SEP)
cb MNA BAT BUS A (Pnl 275) - OPEN
cb MNB BAT C (Pnl 275) - CLOSED
cb MNA BAT C (Pnl 275) - OPEN

MN BUS B LOST - ENTRY

BMAG MODE (3) - RATE 1
FDAI SEL - 1
✓FDAI SOURCE - CMC
AUTO RCS SEL (12) - MNA (ONLY IF BUS LOST AFTER SM SEP)
AC INV 3 - MNA
AC INV 3 AC 2 - ON
AC INV 2 AC 2 - OFF
A11 SM PWR SOURCE MNB - OFF (BEFORE CM/SM SEP)
A11 SM PWR SOURCE MNA - MNA (BEFORE CM/SM SEP)
cb MNB BAT BUS B (Pnl 275) - OPEN
cb MNA BAT C (Pnl 275) - CLOSED
cb MNB BAT C (Pnl 275) - OPEN

AC BUS 1 LOST - ENTRY

BMAG MODE (3) - RATE 2
FDAI SEL - 2
✓FDAI SOURCE - CMC
AC INV 1 MNA - OFF
SUIT COMPR - AC 2
ECS GLY PUMP - AC 2
S BD NORM XPNDR - SEC
S BD NORM PWR AMP - SEC

DATE 2/12/73

AC BUS 2 LOST - ENTRY

BMAG MODE (3) - RATE 1

FDAI SEL - 1

✓FDAI SOURCE - CMC

AC INV 2 MNB - OFF

✓SUIT COMPR - AC 1

✓ECS GLY PUMP - AC 1

BAT BUS A LOST - ENTRY

cb SCS B/D ROLL, P&Y (MNA) (3) (Pnl 8)

Before CM/SM SEP - OPEN

After RCS transfer to CM - CLOSE

cb SCS CONTR/AUTO (2) (Pnl 8) - OPEN

(AFTER APEX COVER JET)

cb MNA BAT C (Pnl 275) - CLOSED

cb BAT BUS A BAT A (Pnl 275) - OPEN

(CLOSED AT PYRO BATT CHK)

BAT BUS B LOST - ENTRY

cb SCS B/D ROLL, P&Y (MNB) (3) (Pnl 8)

Before CM/SM SEP - OPEN

After RCS transfer to CM - CLOSE

cb SCS CONTR/AUTO (2) (Pnl 8) - OPEN

(AFTER APEX COVER JET)

cb MNB BAT C (Pnl 275) - CLOSED

cb BAT BUS B BAT B (Pnl 275) - OPEN

(CLOSED AT PYRO BATT CHK)

2/12/73

DATE

ALL SM PWR SOURCES DISCONNECTED - POWERED FLT
ATTEMPT SM PWR SOURCE RECONNECT (ONE BUS AT A TIME)

IF RECONNECT NOT SUCCESSFUL

IF FC

SM PWR SOURCE 1 - MNB

SM PWR SOURCE 3 - MNA

IF DESCENT BATTS

SM PWR SOURCE 1 - MNB

SM PWR SOURCE 3 - MNA

IF NO SUCCESS

SM PWR SOURCE 2 - MNA & MNB

IF STILL NO SUCCESS

SCE PWR - AUX

EDS AUTO/OFF - OFF

cb MNA BAT C (Pnl 275) - CLOSED

cb MNB BAT C (Pnl 275) - CLOSED

AC BUS OVERLD + AC BUS + MN BUS UNDER V LITES
AFFECTED AC BUS - OFF (REASON - AC BUS SHORT)

FC 1 (3) LITE

VERIFY FC 1 (3) REAC tb - gray

IF tb BP

FC 1 (3) REAC vlv - OPEN (up)

IF tb STILL BP & REAC FLOW ~0

OPEN CIRCUIT FC 1 (3)

SM PWR SOURCE 1 (3) - BAT

2/12/73

DATE

EMER
1-17

SM RCS

SM RCS THRUSTER FAILED ON

BMAG MODE (3) - RATE 2
CHG TO OTHER SC CONT MODE
ROT CONT PWR DIR (2) - MNA/MNB
STOP SPACECRAFT RATES WITH DIRECT RCS
AUTO RCS SEL (16) - OFF

IF CONDITION PERSISTS

AUTO RCS SEL (16) - ON (AS REQ'D)
MAN ATT (3) - ACCEL CMD
STOP SPACECRAFT RATES
cb SCS DIR ULL (2)(Pn1 8) - open
ROT CONT PWR DIR (2) - OFF

IF CONDITION PERSISTS

NEUTRALIZE RHC
SM RCS PRPLNT (AFFECTED QUAD) - OFF

SM RCS LITE

SM RCS HE (2) - CLOSE
SEE RCS 1

SM RCS QUAD SECURE

SM RCS He 1 & 2 (AFFECTED QUAD) (2) - CLOSE
SM RCS PRIM PRPLNT (AFFECTED QUAD) - CLOSE
Fire one jet in affected quad - 2 sec continuously
AUTO RCS SELECT (AFFECTED QUAD) (4)-OFF (except BOOST)

DATE 2/12/73

CM RCS FAILS TO PRESSURIZE OR FEED PRPLNT

IF NO PRESSURIZATION

✓cb EPS BAT BUS (2) (Pn1 229) - CLOSE
✓cb PYRO BUS A/B PYRO BAT A/B (2) (Pn1 229) - CLOSE
✓cb SECS ARM (2) (Pn1 8) - CLOSE
✓SECS PYRO ARM (2) - ARM
✓SECS LOGIC (2) - ON
CM RCS - PRESS

IF NO RCS PRPLNT FEED

✓cb EPS GRP 1 & 3 (Pn1 229) - CLOSE
✓cb SM RCS HTR A&B (Pn1 8) - CLOSE
✓cb RCS PRPLNT ISOL (2) (Pn1 8) - CLOSE
CM RCS PRPLNT - ON

IF STILL NO FEED

cb EPS GRP 5 (Pn1 229) - CLOSE
cb RCS LOGIC (2) (Pn1 8) - CLOSE
CM RCS LOGIC - ON
CM PRPLNT - DUMP MOMENTARILY, THEN OFF

CM RCS

DATE 2/12/73

V05 N09 ALARM CODES

- 00107 Star tracker angles out of limits (P55)
00110 Mark reject has been entered but
ignored
Continue
00113 No inbits (chan 16)
Continue; if alarm recurs use MDC DSKY.
00114 More marks made than desired
Continue
00115 V41 N91 keyed with OPTICS MODE not
in CMC
OPTICS MODE - CMC and OPTICS ZERO - OFF
00116 Optics switch altered before 15 sec
zero time elapsed
OPTICS ZERO - ZERO (15 sec).
00117 V41 N91 keyed but CMC has reserved
OCDU (from start of gimbal test in
P40 until termination of TVC
functional allocation of the
"optics" CDU Driving Output)
Alarm should not occur. Consult STDN.
00120 Optics torque has been requested
but optics have not been zeroed
since last FRESH START or RESTART
OPTICS ZERO - OFF then ZERO (15 sec).
00121 In 0.05 sec following mark, an ICDU
changed by more than 0.033°
Repeat MK.
(m)00205 PIPA saturated
Use SCS control (G&N 12).
00206 The IMU zero routine has been
entered with both the GMBL LOCK
lt and NO ATT lt on
Coarse align to 0,0,0 Reselect V40E
(m)00207 ISS turn-on request not present for
90 sec (G&N 7a)
(m)00210 The IMU is not operating (G&N 12)

DATE 4/3/73

ALARM CODES

- (m)00211 Coarse align error $>2^\circ$
If P52(4) in progress record gyro
torquing angles and perform fine
align check.
Otherwise, see G/2-17 (G&N 12).
- (m)00212 PIPA fail, but PIPA is not being used
PIPA check (G&N 6/7).
- (m)00213 IMU not operating with turn-on request
(G&N 12a)
- 00214 Program using IMU when turned OFF
Exit program.
- (m)00217 IMU coarse align or pulse torque
difficulty has occurred
If code 211 also, perform
211 cure only
Reinitiate current program.
If alarm recurs, terminate use of
ISS (G&N 12).
- 00220 IMU orientation unknown
Align or if aligned set REFSMFLG.
- 00401 Desired middle gimbal angle is excessive
Call N22 - maneuver if MGA $< 85^\circ$ or
realign IMU.
- 00402 Second MINKEY pulse torque must be done
- 00404 Target out of view (90 deg test)
- 00405 Acceptable star pair is not available
- 00406 Rend navigation not operating
Select P20 (Opt. 0 or 4) or continue.
- 00421 W-matrix overflow
Notify STDN but continue.
W-matrix automatically reinitialized at
next mark.
- 00500 Not Enough Jets for Pitch/Yaw (Docked)
- 00501 Not Enough Jets for Roll (Docked)
- 00600 Failure in Phase Match Iteration (P31,32)
- 00601 Failure in Height Maneuver Iteration (P31,32)
- 00602 Failure in Outer Loop Iteration (P31,32)
- 00603 Failure in QRDTPI Iteration (P31,32,33)
- 00611 No TIG for ELEV angle given
- (m)00777 ISS warning caused by PIPA fail
CMC Recovery G/1-23. (G&N 6)
- 01102 CMC self test error

DATE 4/3/73

- (m)01105 Downlink too fast
Rset. If alarm recurs DOWNLINK FAILURE.
(G&N 12).
- (m)01106 Uplink too fast
Rset. If alarm recurs UPLINK FAILURE.
(G&N 12).
- (m)01107 Phase table failure-assume erasable
memory is destroyed. (SSR-3)
If Comm: 1. Contact STDN
2. V74E (erasable dump downlink)
(42 sec - HBR)
3. Do P27 (as necessary)
4. V37E 51E, PRO, V37E00E
5. V46E (V45E docked)
6. OPTICS ZERO - OFF, ZERO
7. Reasonableness check (G/1-23)
If alarm recurs, CMC Failure
- 01301 Arcsin or arccos input is greater than one
Notify STDN, continue.
- (m)01407 VG increasing
(G&N 12).
- 01426 IMU unsatisfactory
Realign or use SCS.
- 01427 IMU reversed
Note FDAI operation is inverted.
- 01520 V37 request not permitted at this time
Wait till COMP ACTY lt.
not on continuously - reselect V37 or if
P62-67, select P00 and then desired program.
- 01600 Overflow in drift test
This is gnd test alarm only.
- 01601 Bad IMU torque abort
See 01600
- 01703 Insufficient time for integration.
TIG slipped
- (m)03777 ISS warning caused by ICDU fail
(G&N 6)
- (m)04777 ISS warning caused by ICDU & PIPA fail
(G&N 6)
- (m)07777 ISS warning caused by IMU fail
(G&N 6)

DATE 2/12/73

- (m)10777 ISS warning caused by IMU & PIPA fail
(G&N 6)
- (m)13777 ISS warning caused by IMU & ICDU fail
(G&N 6)
- (m)14777 ISS warning caused by IMU,ICDU & PIPA
fail
(G&N 6)
- **20430 Orbital integration has been
terminated to avoid possible
infinite loop.
Notify STDN.
Probable S.V. uplink required
- **20607 No solution to conic subroutine
Reselect program.
- **21204 Negative or zero time waitlist call.
If ave-g or ext. vb. on, continue.
Otherwise reselect program.
- **21206 Second job attempts to go to sleep via
keyboard and display program
See 21204.
- **21210 Second attempt is made to stall
Reselect program
Do not attempt use of IMU while CMC is
using it.
- **21302 SQRT called with negative argument
See 21204
- **21501 Keyboard and display alarm during
internal use
See 21204
- **21502 Illegal flashing display
See 21204
- **21521 P01 selected and P11 has already been
performed
Select correct program
- *31104 Delay routine busy
Reselect extended verb or continue with
program.
- *31201 Executive overflow - no vac area
Reselect Extended Verb and/or Continue
Program.
- *31202 Executive overflow - no core sets
See 31201

- *31203 Waitlist overflow - too many tasks
See 31201
 - *31211 Illegal interrupt of extended verb
Reselect extended verb after optics
marking is completed.
 - *32000 Docked DAP cycles overlap.
(m) - Malf procedure indicated
 - ** (2xxxx) - Generates restart (no lt), F37 (POOD00)
 - * (3xxxx) - Restart (no lt) and program
continues (i.e. attempted
recovery) (BAILOUT)
- NOTE - All **alarms act as *type if
they occur when Ave-g is on
or display type extended
verb is active

DATE 2/12/73

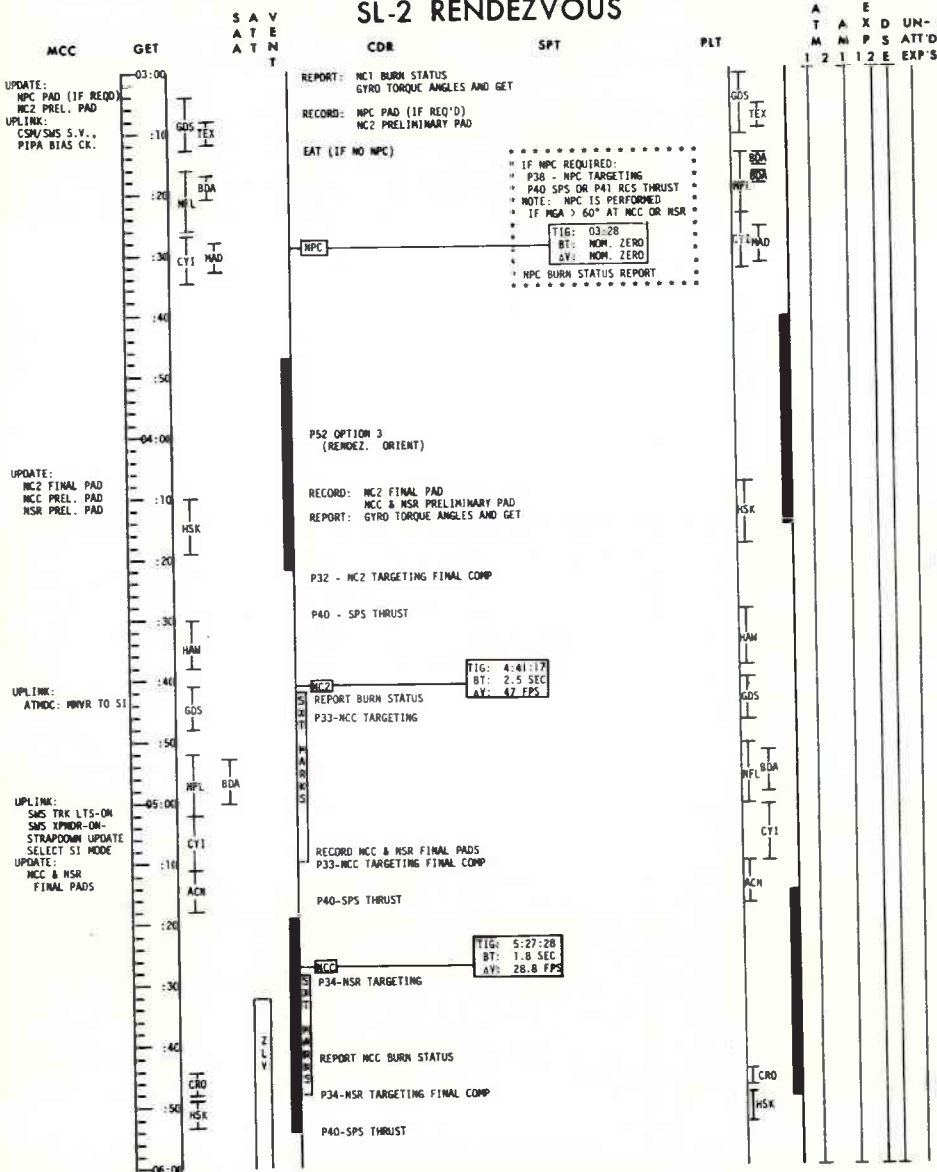


SEVA
L/b

FLIGHT PLAN

GET	GMT	MD/DOY	HOUSTON DATE	REV	BETA	MOON PHASE
03:00 - 06:00	16:00 - 19:00	1/145	MAY 25, 1973			

SL-2 RENDEZVOUS



5/21/73

DATE

SEVA
L/c

FLIGHT PLAN

GET	GMT	MD/DOY	HOUSTON DATE	REV	BETA 4	MOON PHASE
06:00 - 09:00	19:00 - 22:00	1/145	MAY 25, 1973			

SL-2 RENDEZVOUS/DOCKING

MCC

GET

S
A
T
E
N
T

CDR

SPT

PLT

RCDR

A
T
M
2
E
A
M
1
X
D
P
1
UN-
ATT'D
EXP'S

UPDATE:
TP1 PREL PAD &
DOCK ATT

UPDATE:
TP1 FINAL PAD
UPLINK:
MDA SPEC - OFF
OWS N₂ PRESS

UPLINK:
SMS TRK LTS-OFF

UPLINK:
SMS ATT HOLD
MODE - TACS
SMS MNVR TO FLY
AROUND ATT

UPLINK:
SMS N2 PRESS
SMS DOCK LTS-ON
UPDATE:
GO/NO GO FOR
SOFT DOCK
SUNSET TIME

UPLINK:
SMS DOCK LTS-OFF
(IF DOCKED)

06:00
10 HAM
20 GDS
30 BDA NFL MIL
40 ACN
50
07:00
10
20 CRO
30 GNM
40
50
08:00
10 GDS TEX MIL
20 BDA
30
40
50
09:00
10 CRO

NSR
MNVR TO HEADS UP ATTITUDE
REPORT NSR BURN STATUS
MNVR TO ACQUIRE SMS (HEADS UP)

TIG: 6:04:28
BT: 0.7 SEC
AV: 18.8 FPS

RECORD TP1 PRELIMINARY PAD/DOCK ATTITUDE
BACKUP CHART MEASUREMENTS ARE TAKEN AT
TP1-32, -28, -24, AND -16
RECORD FINAL TP1 PAD

P35-TP1 TARGETING FINAL COMP

P40 SPS THRUST

TP1
P36-TPM TARGETING
BACKUP CHART MEAS. TAKEN AT
TP1 + 4:30, + 7:30 and + 8:30
P41 - RCS THRUST

TIG: 07:02
BT: 0.7 SEC
AV: 20.6 FPS

TPM1
P36-TPM TARGETING
BACKUP CHARTS MEAS. TAKEN AT TP1 +16:30, +19:30, and +20:30
P41 - RCS THRUST

TPM 2
P37 RENDEZ. FINAL PHASE
PRE-BRAKING CHECKLIST
DEACT PRIMARY AND SECONDARY EVAPORATORS
P48 - BRAKING
OWS PHOTOS
STATION KEEP

FLY AROUND

CAPTURE LATCH DOCKING

EVA CHECKLIST

SUPPER

MISSION	EDITION	PUBLICATION DATE
SL-2	FINAL, REV B	MAY 21, 1973

FLIGHT PLANNING BRANCH

S	SOL	A-SOL
A		

5/23/73
5/23/73
DATE

SEVA
L/d

FLIGHT PLAN

GET	GMT	MD/DOY	HOUSTON DATE	REV	BETA 4	MOON PHASE
09:00 - 12:00	22:00 - 01:00	7/145	MAY 25, 1973			

DOCKING

MCC

GET

S
A
V
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N
T

CDR

SPT

PLT

RCOR
E
A
T
A
X
D
UN-
M
P
S
ATT'D
1
2
E
EXP'S



SEVA PREP

SEQUENTIAL LOGIC CHECK

GO/NO GO FOR SEVA

CAPTURE LATCH UNDOCK

SEVA

CLOSE HATCH
CABIN REPRESS

PREDOCK CHECKLIST

DOCK DOCKING CHECKLIST

SMS ACT CHECKLIST

CM/MDA TUNNEL PRESS INTEGRITY CHECK [1-1]
BAT A CHANGE [1-1]
CN TUNNEL HATCH REMOVAL [1-2]
DOCKING LATCH VERIFICATION [1-2]
CN HATCH INSTALLATION [1-3]
SM RCS PROPLANT RECONFIG [1-3]
CN HATCH INTEGRITY CHECK [1-4]

DATE 5/21/73

MISSION	EDITION	PUBLICATION DATE
SL-2	FINAL, REV 8	MAY 21, 1973

FLIGHT PLANNING BRANCH

S A L	SOL	A-SOL

SEVA
L/e

FLIGHT PLAN

GET	GMT	MD/DOY	HOUSTON DATE	REV	BETA 4	MOON PHASE
0:12:00/0:15:00	01:00 - 04:00	1/146	MAY 25, 1973			

RCDR
A E
T A X D UN-
M M P S ATT'D
1 2 1 1 2 E EXP'S

MCC
GMT
TIME SCALE IN GMT
(12:00)

CDR SPT PLT

UPLINK:
APCS GAIN CHANGE
MDA SPEC - ON
(MATT ONE RTR)
MDA SPEC -
CMT/THRESH

HELMETS, GLOVES, & PGA'S DOFFING & STORAGE

(SEVA L/1-13)

AT GET 13:00 SET MISSION TIMER
TO 02:00 GMT

SEVA DEBRIEFING

Perform Quiescent Panel Config.
(S/3-1 thru 3-9) Except:
CM 326: Surge TK 02 VLV-OFF (S/3-1)
380: All Items (S/3-1)
2: POT H2O HTR-OFF (S/3-5)
2: Suit Ckt H2O Accum Auto-Off
(Center) (S/3-5)
4: Suit Compr 1-AC1 (S/3-7)
275: Delete All Items (S/3-7)
Do Not Do 3-10 and 3-11

DINNER [1-5]

REPORT: DOCKING TUNNEL INDEX ANGLE

CM 325: CABIN PRESS RELF
VLV (2) - CLOSE
CM 375: DELETE ALL ITEMS

DO

CM 399: AUX Gx4 EVAP
IN TEMP VLV - MAX

MISSION	EDITION	PUBLICATION DATE
SL-2	FINAL REV B	MAY 21, 1973

SOL	A-SOL

FLIGHT PLANNING BRANCH

5/21/73

DATE

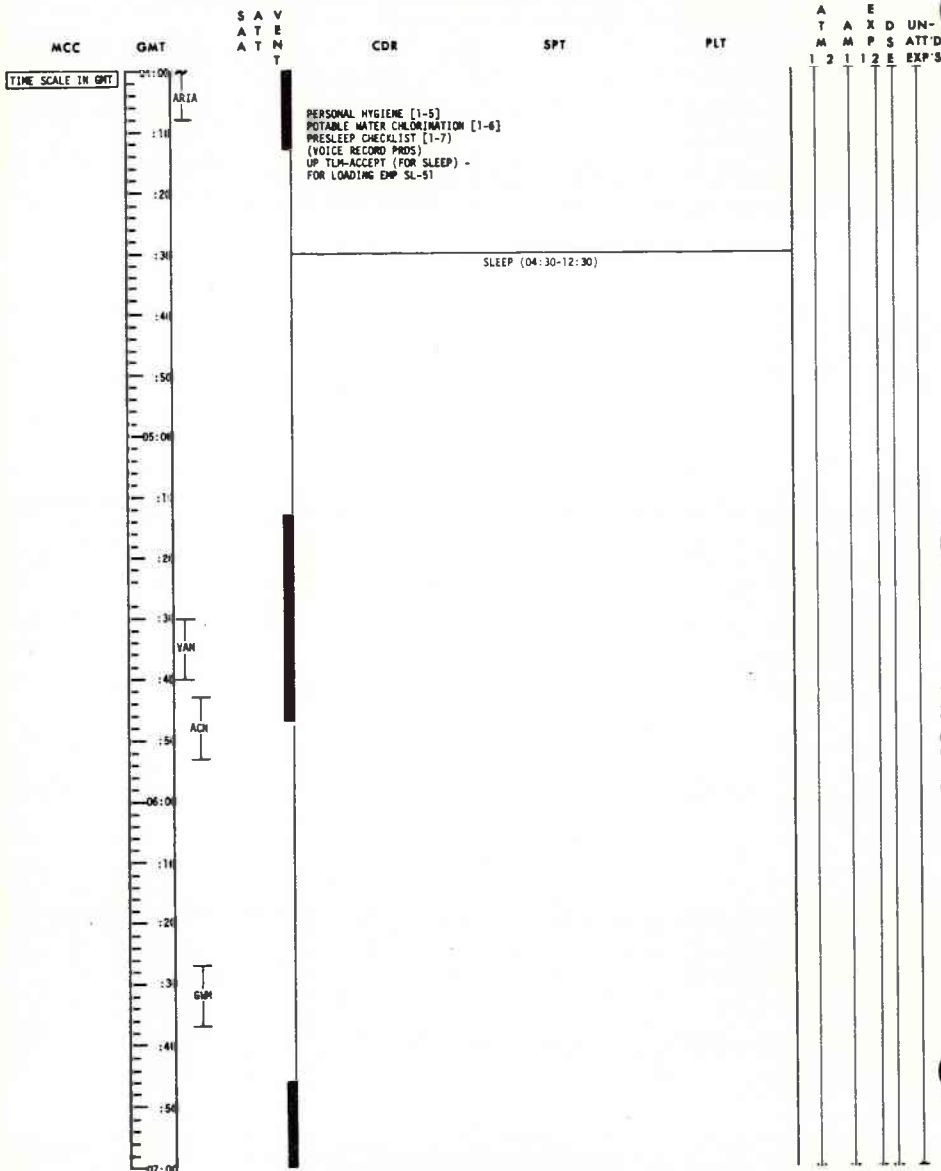
5/23/73 (P+I)

add
WRT
3-11

SEVA
L/f

FLIGHT PLAN

GET	GMT	MO/DOY	HOUSTON DATE	REV	BETA	MOON	
15:00 - 18:00	04:00 - 07:00	2/146	MAY 26, 1973			PHASE	



MISSION	EDITION	PUBLICATION DATE
SL-2	FINAL REV A	MAY 21, 1973

FLIGHT PLANNING BRANCH

SOL	A-SOL
---	---

5/21/73
DATE

STANDUP EVA (SEVA)

SEVA CHECKLIST

SEVA CONTROL CONFIG

DAP LOAD:

R1 10102

R2 11111

AUTO RCS SELECT A3 - OFF (all
others - MNA)

cb SCS A/C ROLL MNB - open

B/D ROLL MNB - open

PITCH MNB - open

YAW MNB - open

1 ROT CONTR PWR NORMAL 1 - OFF

ROT CONTR PWR NORMAL 2 - AC/DC

ROT CONTR PWR DIRECT 1 - OFF

ROT CONTR PWR DIRECT 2 - OFF

(MNA/MNB if reqd)

cb SCS 2 CONTR/DIRECT MNA - OPEN

S/C CONT - CMC

CMC MODE - FREE

RATE - ~~HIGH~~ LOW

DEADBAND - MIN

BMAG (3) - RATE 2

MANUAL ATTITUDE sw (3) - RATE CMD

LOGIC SEQUENCE CHECK

SECS LOGIC (2) - OFF (verify)

SECS PYRO ARM (2) - SAFE (verify)

cb SECS LOGIC (2) - close (verify)

cb SECS ARM (2) - close

**After STDN GO - SECS LOGIC (2) -
ON (up)

Receive STDN GO/NO-GO for PYRO ARM

DATE 5/24/73

5/24/73

SEVA CHECKLIST

CABIN PREP FOR SEVA

CDR

379 Adjust PRIM ACCU QTY, PRIM ACCUM
FIL vlv - ON until 50-55%, then
OFF

Unstow 3 TSBs (U1) place as
desired

Fill drink bags as desired:

Fill one hour before install

Use water gun in short port

Fill approx 8 oz.

Let sit, bubbles will concentrate

Work bubble to drink port

Bite and suck out gas

Unstow SEVA cue card and SWS map and
SAS from CSM DATA CARD KIT, secure

Stow optics

1

ROT CONT PWR NORMAL 1 - OFF (Ver)

ROT CONT PWR DIRECT 1 - OFF (Ver)

RHC NO. 1 lock & stow (F2)

Lower RHC NO. 1 strut and lock

mount push locks

Relocate LEB stowage as required
for center couch removal

Remove center couch

Close and lock clamps

Stow couch under CDR couch

using 2 straps (R5)

Unstow tool heads (A2) prep as reqd

Configure 3 pole segments

Install meter covers (A4)

Unstow Tool Kit (U3)

Remove jack screws, verify full
open, secure as desired

Snap Tool Kit to girth ring be-
hind CDR couch

SEVA CHECKLIST

5/23/73 (R+I)
5/24/73

DATE

SEVA
L/1-3

SIDE HATCH PREP

GN2 vlv handle - PULL TO VENT
Gage reads - min
Leave in vent position
Verify/install alignment mark on
counterbalance
Disconnect hatch counterbalance
lock pin (Tool B, turn ccw to
stop)

EQUIPMENT PREP FOR SEVA

Unstow EVA eqpt bag (A1) snap
in LEB
Unstow SPT EV gloves R13, place
in TSB top pocket
Unstow PGA Maint Kit (R13)
Clean & anti fog 3 helmets per
instruction sheet, tissue in B1
Stow PGA Maint Kit in TSB
Stow PLT accessory bag (with IV
gloves) in TSB
Stow CDR & SPT helmets in
helmet bags

Unstow SEVA, install on PLT helmet,
place in tunnel
Verify pen lights available
Install 3 MDC bars (EVA bag)
Secure utility pole tether (EVA
bag/35 MM eqpt bag) to LH MDC
bar

Secure waist tether (EVA bag/
35MM eqpt bag) on MDC monkey
bar for hand hold - LOCK

Install DAC in RH window, Mag CX02
(B8), T/8, 1/250, 15ft, 1 FPS,
test

DATE 5/21/73
5/23/73 (P+I)

SEVA
L/1-4

UMBILICAL CONNECTIONS

ALL

6/9/10

SUIT PWR - OFF

PWR - OFF

AUDIO CONT - NORM (verify)

PLT Connect to pnl 300 long hose,
route as reqd

PLT Attach umbilical tether to PGA
D-ring, lock & install pin,
verify tether secure at pnl 300

PGA 02 hoses R/R & B/B (verify)

300/301/302

SUIT FLOW vlv - FULL FLOW (verify)

6/9/10

PWR - AUDIO/TONE

SUIT PWR - on (up)

CONFIGURE COMM

6/9/10

MODE - VOX

VOX SENS - as reqd (~6)

S-BD - T/R (verify)

AUDIO CONT - NORM (verify)

INTERCOM - T/R (verify)

SYSTEMS PREP FOR DEPRESS

CDR

325 CAB PRESS RELF vlv (2) - NORM

326 REPRESS PKG vlv - FILL

SURGE TK vlv - ON (verify)

351 CAB REPRESS vlv - OFF

600 EMER 02 vlv - CLOSED (verify)

601 REPRESS 02 vlv - CLOSED (verify)

602 REPRESS 02 ind - approx 900 psia

2 SURGE TK PRESS - approx 900 psia

2 CAB FANS (2) - OFF

2 02 HTRS (2) - AUTO (verify)

DATE 5/21/73

SEVA
L/1-5

- 226 cb CYRO 02 100W HTR 1 MNA - close
(verify)
cb CYRO 02 100W HTR 2 MNB - close
(verify)
cb CYRO 02 50W HTR 1 MNB - ~~close~~ ~~open~~ ~~close~~
(verify)
cb CYRO 02 50W HTR 2 MNA - ~~close~~ ~~open~~ ~~close~~ (verify)
(verify)

ALL Verify gas connector plugs
installed and all connectors
locked
PGA diverter vlv - horiz
Install drink bags
Don wristlets & comfort gloves
(PGA pkt)
Configure couch pans as reqd

- Go to SEVA CUE CARD -

- PLT Don helmet/SEVA (do not rotate
helmet after attachment)
Lower SEVA protective visor
Don EV gloves and cover gloves
(EVA bag)
Don wrist tether (remove 2 from
EVA bag/35MM eqpt bag)
SPT Don helmet
Don EV gloves (TSB)
Don wrist tether
CDR Don helmet
Don visor
Don IV gloves

ALL Verify helmet & gloves locked

DATE 5/21/73
5/23/73 (P+I)

PURGE SUIT CIRCUIT

ALL

- 7 DIRECT 02 vlv - OPEN (ccw) for
1 minute
- 2 02 FLOW ind - 1.0 lb/hr (pegged)
02 FLOW HI lt - on
M/A - on, reset
- 7 DIRECT 02 vlv - close (cw)
- 2 02 FLOW HI lt - out
02 FLOW ind - 0.2 lb/hr

INTEGRITY CHECK

ALL

- 380 SUIT CKT RETURN VLV - close (push)
- 2 SUIT PRESS ind - 4.7-5.3 psia
02 FLOW ind - 0.2-0.4 lb/hr

CAUTION

SUIT TEST vlv should remain in the PRESS position until suit circuit pressure is stabilized to preclude seal scarring. If repositioning of SUIT TEST vlv from PRESS is required prior to suit pressure and 02 flow stabilization, perform the following:

1. 02 DEMAND REG vlv - OFF
2. Allow 15 sec (min) stabilization time
3. Reposition SUIT TEST vlv - DEPRESS or OFF as applicable
4. When suit pressure stabilized, 02 DEMAND REG vlv - BOTH

5/21/73

DATE

SEVA
L/1-7

- 380 SUIT TEST vlv - PRESS
DIR 02 - OPEN
2 02 FLOW ind - 1.0 lb/hr (pegged)
2 02 FLOW HI lt - on
M/A - on, reset

When SUIT PRESS ind 1.5-2.0 psi
greater than CAB PRESS, SUIT CKT
RETURN VLV - open then close

When SUIT PRESS ind 3.5 to 4.0 >CAB
PRESS, DIRECT 02 - CLOSE

- 2 SUIT PRESS ind - 8.8-9.8 psia
Cuff gage - 4.1-4.5 psig
2 02 FLOW HI lt - out

Allow 02 FLOW to stabilize 15 sec
02 FLOW shall remain below 0.80
lb/hr for 30 sec after stabiliza-
tion

- * If 02 FLOW greater than *
* 0.80 lb/hr reverify all con- *
* nections and repeat INTEGRITY*
* CHECK. Continue after re- *
* check if 02 FLOW less than *
* 0.97 lb/hr for 30 sec after *
* stabilization *

- 380 SUIT TEST vlv - DEPRESS
2 02 FLOW ind - 0.2-0.4 lb/hr
SUIT PRESS ind - slightly greater
than CABIN PRESS ind
380 SUIT TEST vlv - OFF
02 DEMAND REG vlv - BOTH (verify)

5/21/73

DATE

UNDocking

- THC & RHC - UNLOCKED ——— THC - ON
- 8 cb DOCK PROBE (2) - close
- 2 PROBE EXT D/REL - EXT D/REL (HOLD)
(<20 sec)
- After approx 2 sec -X translation
& probe disengages, PROBE EXT D/
REL - RETRACT/OFF - t.b.(2)-gray
- 8 cb DOCK PROBE (2) - open
- CMC mode - AUTO
- Maneuver to SEVA stationkeeping
position
- BMAGs (3) - ATT 1/RATE 2

CM DEPRESS

- SPT
- 351 EMER CAB PRESS sel - OFF
- SIDE HATCH DUMP vlv - open, 02 HI
FLOW warning lt may come on prior
to cabin press reg lock up
- 2 Monitor CAB PRESS ind to 3.0-3.25
psig, then SIDE HATCH DUMP vlv -
close
- 380 SUIT CKT RETURN VLV - open, then
close
- 02 FLOW ind <0.5 lb/hr (nominal
0.24 lb/hr)
- 2 Verify SUIT PRESS 3.5-4.0 psia
- SIDE HATCH DUMP vlv - open
- 2 Monitor CAB PRESS to 0.0 psia
Verify stable SUIT PRESS 3.5-4.0
psia and HI 02 FLOW lt - off (02
FLOW ind <0.5 lb/hr)

5/23/73 (P+I)
5/24/73

DATE

SEVA
L/1-9

HATCH OPENING

Lock pin release knob - unlock
Lock pin indicator released (white
to yellow)

Gear box sel - unlatch
Actr handle sel-U

*** IF SM RCS THRUSTER FAILS ON: ***

Unstow ACTR handle

*** SC CONT - SCS**

Unlock hatch

*** AUTO RCS (AFFECTED JET) - OFF ***

ACTR handle - SEL-L

*** (ODD JET / ROTATION) ***

Stow actr handle

*** DAMP RATES WITH ***

Open hatch

*** RHC ***

Verify hex clears

Verify hatch full open

SIDE HATCH Dump vlv - CLOSE

SEVA OPERATIONS

PLT & CDR Lower SEVA visors

DAC - ON

Perform tasks with utility pole

DEPLOY SAIL IF REQD PER SEVA SAIL
DEPLOYMENT CHE CARD

CM POST SEVA

BMAG (3) - RATE 2

Maneuver CSM to POST SEVA position

HATCH CLOSING

Verify hatch clear of obstructions

Unstow lanyard

Pull and close hatch

Verify hex clears

Close hatch

DATE

5/23/73
5/27/73
5/24/73

Gear box SEL - latch
Unstow ACTR handle
Lock hatch

Verify lock pin dropped in (white
to white)
Stow ACTR handle
ACTR handle select-N
Gear box SEL-latch (verify)

*Hatch will not latch (frozen *
* gearbox) - *
* Install (3) jackscrews to re- *
* strain hatch in closed position *
* Use tool B to remove (2) clevis *
* pins connecting linkage to *
* gearbox and (1) clevis pin from *
* linkage in ~~corner above gearbox~~ *
* Tighten jackscrews to close hatch *
* as far as possible *
* Use tool F on flats of latch *
* bellcrank to drive latch to *
* over-center closed position *
* (Apply tool F to upper ²latch on *
* hinge side ~~of~~ drive the lower *
* and hinge side linkage closed). *
* Apply tool F to center latch to *
* drive upper linkage closed. *
* Gearbox side linkage may not *
* close if gearbox is in full *
* open position. *
* Install clevis pins in threaded *
* hole in linkage bell cranks at *
* upper ~~gearbox side~~ (Clevis pin *
* installed when approx half the *
* threads are visible). *

Upper LH CRNR

LH CRNR¹ LWR
RH CRNR

5/23/73 (P+L)
5/21/73

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CABIN REPRESSURIZATION

SIDE HATCH DUMP vlv - CLOSE (ver)
325 CAB PRESS REL vlv (2) - NORMAL
(latch on) (verify)
2 02 PRESS IND - SURGE TK (verify)

SEVA
L/1-11

601 REPRESS 02 VLV - OPEN then CLOSE
at cabin press of 1 psia

Monitor cabin press ind for 30 sec
for gross leakage

601 REPRESS 02 vlv - OPEN
When SURGE TANK PRESS decreases
to 150 psi, pnl 326 REPRESS
PKG vlv - OFF

2 CAB PRESS ind - ~3.0 psia (1 min)
351 CAB REPRESS vlv - OPEN (cw), adjust
to maintain >150 psia in SURGE TK
602 REPRESS 02 PRESS ind - ~0 psia
601 REPRESS 02 vlv - CLOSE

2 CAB PRESS - 4.7-5.3 psia
351 CAB REPRESS vlv - OFF
When SURGE TK >400 psi, pnl 326
REPRESS PKG vlv - FILL
351 EMER CABIN PRESSURE sel - BOTH

DAC - OFF

- Go to SEVA L/1-11

POST SEVA CONTROL CONFIG

AUTO RCS SELECT - A3 - MNA
RELOAD DAP R1 11102

Translate to docking position and
maneuver to docking attitude

DATE 5/21/73

DOCKING ATTITUDE

NOMINAL

UPDATE

N22	R	+				0	0		+				0	0
	P	+				0	0		+				0	0
	Y	+				0	0		+				0	0

PRE-DOCKING SWITCH LIST

BMAG MODE(3)-ATT 1/RATE 2	DOCKING PROBE EXT/REL tb(2)-gray
cb DOCK PROBE(2)-CLOSE	SPOT LIGHT-ON(down)
DOCKING PROBE RETRACT(2)-OFF(ctr)	SECS PYRO ARM(2)-ON(up)
DOCKING PROBE EXT/REL-RETRACT	FC REACS VALVES-LATCH (verify)

(11111) V44E (Verify docked DAP
(00146) DATA LOAD (N89=0.2000,
(00030) 000.50)

DOCKING CHECKLIST

TO INITIATE CAPTURE

VERIFY STATION KEEPING ON SWS +X AXIS
(RANGE=40 TO 100 FT) (DOCK TGT=1.8 TO 0.8° IN COAST)
LOAD EMS WITH 100.0 FPS
EMS FUNC/MODE-ΔV/NORMAL
DAC-ON
THC-INITIATE 0.4 ±0.1 FPS CLOSING RATE
RHC-MAINTAIN MINIMUM RELATIVE ALIGNMENT ANGLES

S/S 15-3

AT CAPTURE (IF NO CAPTURE-SEE BACKUP DOCKING PROCEDURES (S/2-1))

DOCKING PROBE EXT/REL tb(either)-bp (IF GRAY-SEE DOCKING MALFUNCTION PROCEDURES #2, Pg 13-3 IN CSM MALFUNCTION PROCEDURES BOOK)

CMC MODE-FREE
RHC PWR DIRECT(2)-OFF(ctr)
REPORT CAPTURE TO STDN
ALLOW PROBE TO DAMP SC MOTION (10 SEC)
THC-NULP PITCH AND YAW MISALIGNMENT<3°
RHC-NULP ROLL MISALIGNMENT<5°
DOCKING PROBE RETRACT PRIM-1 (SEC-2 IF REQ)

AT DOCK LATCH

DOCKING PROBE EXT/REL tb(2)-gray

AFTER HARD DOCK

V45E	EMS FUNC/MODE-OFF/STBY
SECS PYRO ARM(2)-SAFE	DBD/RATE-MIN/LOW
SECS LOGIC(2)-OFF	THC PWR-OFF
cb SECS ARM(2)-OPEN	RHC PWR NORMAL#1-OFF(ctr)
cb DOCK PROBE(2)-OPEN	RHC PWR DIRECT(2)-OFF(ctr)
DOCKING PROBE EXT/REL-OFF	SC CONT/MODE-CMC/FREE
DOCKING PROBE RETRACT(2)-OFF(ctr)	BMAG MODE(3)-RATE 2
DAC-OFF	AUTO RCS SELECT(A1,D2,A3,C4,B3,D4)-OFF
SPOT LIGHT-OFF(up)	cb SCS A/C & B/D ROLL(4)-OPEN
FC REACS VALVES-NORM	cb SCS PITCH & YAW(4)-OPEN
VHF RANGING-OFF	THC-LOCKED
COAS PWR-OFF	RHC(2)-LOCKED

DATE 5/23/73 (P+I)

SEVA
L/1-13
GO TO SEVA L/d
PGA DOFFING AND STOWAGE

Verify:

- 326 SURGE TK >400 psi
351 REPRESS PKG vlv - FILL
351 EMER CABIN PRESSURE sel - BOTH

- Doff helmets/gloves
380 SUIT CIRCUIT RETURN vlv - OPEN
Stow PRDs, suit pocket equip,
watches & bands (top pocket -
TSB)
Stow comfort gloves & wristlets
(accessory bag)

Unstow 02 umbilical screen caps
(B2)
Install hose screen on return hose

6/9/10

- PWR - OFF
SUIT PWR - OFF
AUDIO CONT - NORM

- Stow CCA (accessory bag)
Doff PGAs (EVA C/L 3.2-2)
Install helmets & IV gloves
on PGAs
Connect electrical dust covers (B2)
to PGAs
Remove center couch from under CDR
couch and reinstall
Stow couch straps R-5
Stow PGAs (under couches)

- Doff UCTAs (when full)
Place UCTA clamps (A6) on UCTAs
Stow UCTAs in WMS bag (A5)
Don UCTAs (A6)

- Stow OBS/biobelts (accessory bags)
Don CWG harnesses and 1t wt head-
sets (U2)

DATE 5/21/73
5/23/73(P+I)

6/9/10

PWR - AUDIO/TONE
SUIT PWR - ON (up)

VERIFY COMM

MODE - INTERCOM/PTT
SBD - T/R (verify)
AUDIO CONT - NORM (verify)
INTERCOM - T/R (verify)
Don trousers, jackets, and boots
(U2)
PRD's (TSB) to jackets
Stow in EVA bag:
MDC bars
Waist tether on MDC
PLT EV gloves/cover gloves
Utility pole tether

Stow EVA bag in A1
Stow tool heads in A2
Stow meter covers in A4
Stow utility poles
Reconfigure RHC #1

Stow SPT EV gloves in R13
Stow PGA Maint Kit (TSB) in R13
Stow SEVA & CDR visor under couches

Stow hatch lanyard & D-ring
Verify hatch counterbalance alignment
Install hatch counterbalance lock
in pin (tool B - turn cw to stop)
GN2 vlv handle - OFF (center)

Stow jack screws in Tool Kit
Stow Tool Kit in U3

Stow DAC & Mag in B3

5/23/73 (P+I)
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SEVA
L/1-15

Verify:

602 REPRESS 02 ind ~900 psia

2 SURGE TK PRESS ~900 psia

226 C/B CYRO O₂ SOW HTR & HNB CLOSE

Refer to SEVA day timeline for app-

licable SWS ACTIVATION C/L pro-

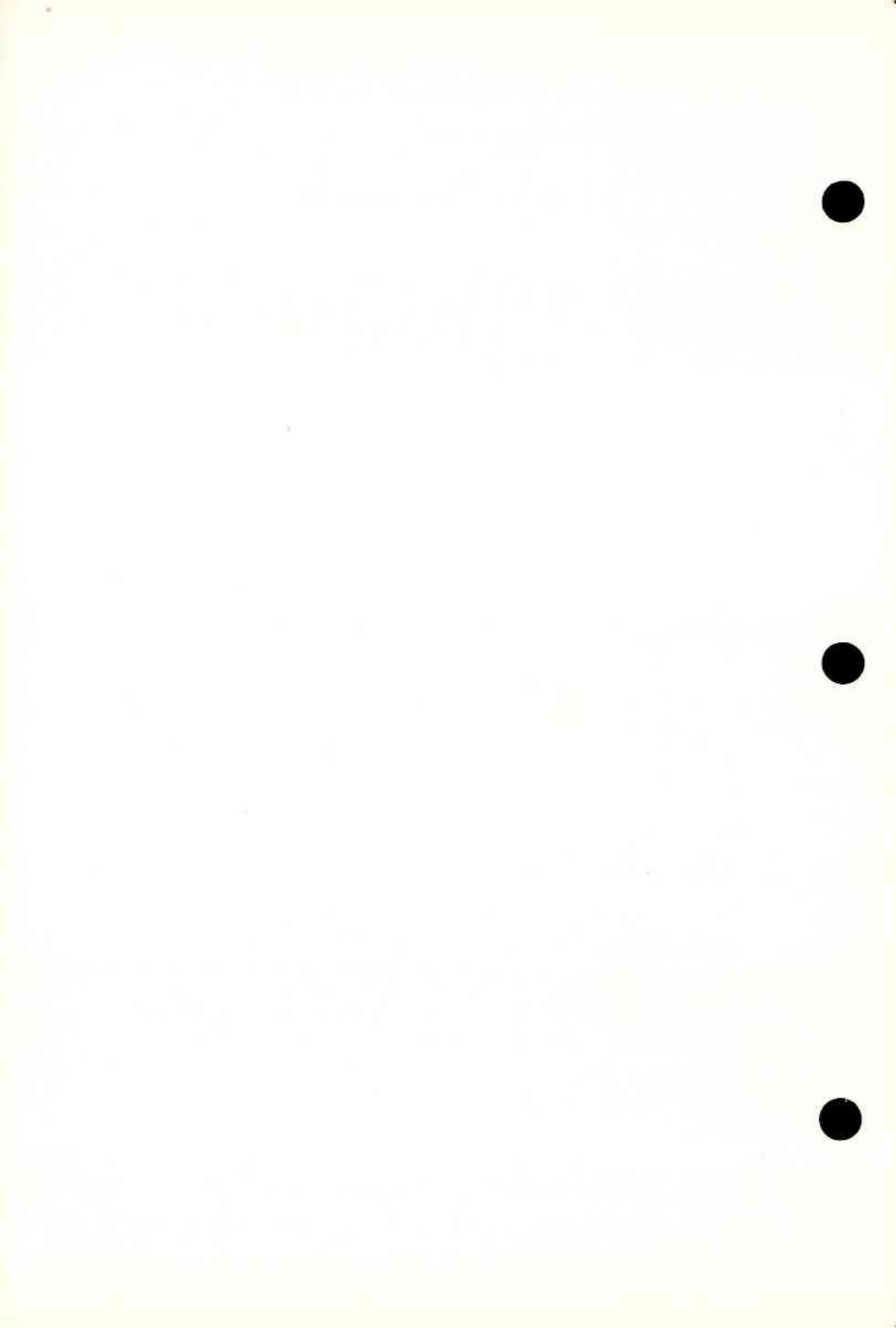
cedures (flight plan pages) (SEVA L/e)

5/24/73

~~5/23/73~~ (P+I)

5/24/73

DATE



BLOCK DATA PAD DESCRIPTION

- a. Block data pads provide a single impulse deorbit capability in the event of communications failure or rapid deorbit requirements
- b. Deorbit burn ignition time (TIG) referenced to GMT in Julian days (i.e., 1 January = day 1)
- c. Deorbit burn velocity change, fps (ΔVC)
- d. Deorbit burn duration (BT)
- e. Backup GDC align attitude; R, P, & Y
- f. GDC burn attitude is 0° , 180° , 0°
- g. .05G time referenced from retroburn (RET .05G)
- h. EMS initialization value for range-to-go at .05G (RTOGO .05G)
- i. EMS initialization value of inertial velocity at .05G (VIO .05G)
- j. Entry interface (400,000-ft altitude) time referenced from retroburn (RET EI)
- k. Target latitude and longitude of splashdown

DATE 5/17/73

SINGLE IMPULSE
(NO COMM)

BLOCK DATA													
PET	TIG DAY	X	X	X	N/A	X	X	X	N/A				
	HR	X	X	X	X	0	1	X	X	X	X	0	4
	MIN	X	X	X	X	2	9	X	X	X	X	0	4
	SEC	X	X	X	X	3	9	X	X	X	X	2	8
	ΔVC	X	X	2	6	1	5	X	X	2	3	6	5
	BT	X	X	0	0	1	2	X	X	0	0	1	1
GDC ALIGN R		X	X	X				X	X	X			
	TW P	X	X	X				X	X	X			
	SETTINGS Y	X	X	X				X	X	X			
BRN ATT GDC FDAI		R, P, Y						0°, 180°, 0°					
	RET .05G	X	X	1	5	2	4	X	X	1	9	2	2
	RTOGO .05G	+	1	7	0	1	7	+	1	0	7	9	4
	VIO .05G	+	2	5	7	6	5	+	2	5	8	2	4
	RET EI	X	X	0	9	3	4	X	X	1	7	0	3
	LAT	+	0	4	8	8	2	+	0	1	1	4	5
	LONG	-	0	1	7	9	5	-	1	6	6	0	0
REMARKS: ① Assumes NO RENDEZVOUS MANEUVERS													
② R=358, P=210, Y=359 (BURN ATTITUDE FOR LAUNCH REFS)													
③ R=359, P=059, Y=359 (BURN ATTITUDE FOR RENDEZVOUS REFS)													
④ ROLL RIGHT 55°/EMS ENTRY													
REMARKS: ① Assumes NO RENDEZVOUS MANEUVERS													
② R=000, P=312, Y=357 (BURN ATTITUDE FOR LAUNCH REFS)													
③ R=000, P=161, Y=358 (BURN ATTITUDE FOR RENDEZVOUS REFS)													
④ ROLL RIGHT 55°/EMS ENTRY													

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SINGLE IMPULSE
(NO COMM)

DATE 5/17/73

BLOCK DATA													
GMT	TIG DAY	X	X	X	1	4	6	X	X	X	1	4	7
	HR	X	X	X	X	0	2	X	X	X	X	0	2
	MIN	X	X	X	X	5	3	X	X	X	X	0	2
	SEC	X	X	X	X	5	7	X	X	X	X	4	6
	Δ VC	X	X	4	3	1	6	X	X	4	3	1	6
	BT	X	X	0	0	1	9	X	X	0	0	1	9
GDC ALIGN R		X	X	X				X	X	X	1	8	8
	TW P	X	X	X				X	X	X	0	6	4
	SETTINGS Y	X	X	X				X	X	X	0	0	1
BRN ATT GDC FDAI		R, P, Y						0°, 180°, 0°					
RET .05G		X	X	2	9	2	3	X	X	2	9	3	3
RTOGO .05G		+	1	2	0	0	3	+	1	1	9	6	4
VIO .05G		+	2	5	9	8	5	+	2	5	9	9	8
RET EI		X	X	2	6	5	6	X	X	2	7	0	2
LAT		+	0	0	9	5	3	+	0	3	1	6	2
LONG		-	1	5	2	1	0	-	1	7	3	0	0
REMARKS: ① GMT UNDOCKING 146:02:23													
② ROLL RIGHT 55°/EMS ENTRY													
③ R=001, P=050, Y=359 (BURN ATTITUDE RENDEZVOUS REF5MMAT)													
REMARKS: ① GMT UNDOCKING 147:01:42 ② ROLL LEFT 55°/EMS ENTRY													
③ GDC ALIGN TW SETTINGS FOR GDC ALIGNMENT TO ATM (Pg 6/349)													
④ IF NOT IN SOLAR INERTIAL:													
A GDC ALIGN WILL BE DONE DURING TIG-3 MIN HORIZ CK.													
B AFTER UNDOCKING TRACK AFT HORIZ ON 13° WINDOW MK (HEADS DN).													

BLOCK DATA

GMT	TIG DAY	X	X	X	1	4	8	X	X	X	1	4	9
	HR	X	X	X	X	0	1	X	X	X	X	0	0
	MIN	X	X	X	X	2	4	X	X	X	X	4	1
	SEC	X	X	X	X	1	7	X	X	X	X	0	0
	ΔVC	X	X	4	3	1	6	X	X	4	3	1	6
	BT	X	X	0	0	1	9	X	X	0	0	1	9
	GDC ALIGN R	X	X	X	1	8	9	X	X	X	1	9	1
	TW P	X	X	X	0	7	3	X	X	X	0	6	5
	SETTINGS Y	X	X	X	0	0	1	X	X	X	0	0	1
BRN ATT GDC FDAI		R, P, Y						0°, 180°, 0°					
	RET .05G	X	X	2	9	2	8	X	X	2	9	1	3
	RTOGO .05G	+	1	2	0	7	6	+	1	1	9	2	9
	VIO .05G	+	2	5	9	9	3	+	2	5	9	9	1
	RET EI	X	X	2	7	0	0	X	X	2	6	4	5
	LAT	+	0	2	3	9	2	+	0	2	3	9	0
	LONG	-	1	5	4	8	3	-	1	5	4	6	5

REMARKS: ① GMT UNDOCKING 148:01:02 ② ROLL RIGHT 55°/EMS ENTRY

③ GDC ALIGN TW SETTINGS FOR GDC ALIGNMENT TO ATM (Pg 6/3-19)

④ IF NOT IN SOLAR INERTIAL:

① GDC ALIGN WILL BE DONE DURING TIG-3 MIN HORIZ CK

② AFTER UNDOCKING TRACK AFT HORIZ ON 13° WINDOW MK (HDS DN)

REMARKS: ① GMT UNDOCKING 149:00:20 ② ROLL LEFT 55°/EMS ENRY

③ GDC ALIGN TW SETTINGS FOR GDC ALIGNMENT TO ATM (Pg 6/3-19)

④ IF NOT IN SOLAR INERTIAL:

① GDC ALIGN WILL BE DONE DURING TIG-3 MIN HORIZ CK

② AFTER UNDOCKING TRACK AFT HORIZ ON 13° WINDOW MK (HDS DN)

DATE 5/3/73

BLOCK DATA

TIG DAY	X	X	X	1	5	0	X	X	X	1	5	1
HR	X	X	X	X	0	1	X	X	X	X	0	0
GMT												
MIN	X	X	X	X	3	4	X	X	X	X	5	4
SEC	X	X	X	X	3	8	X	X	X	X	0	9
ΔVC	X	X	4	3	1	6	X	X	4	3	1	6
BT	X	X	0	0	1	9	X	X	0	0	1	9
GDC ALIGN R	X	X	X	1	9	3	X	X	X	1	9	5
TW P	X	X	X	0	7	2	X	X	X	0	7	6
SETTINGS Y	X	X	X	0	0	1	X	X	X	0	0	1
BRN ATT GDC FDAI	R, P, Y						0°, 180°, 0°					
RET .05G	X	X	2	9	0	9	X	X	2	9	0	6
RTOGO .05G	+	1	2	0	2	4	+	1	2	0	7	6
VIO .05G	+	2	5	9	8	5	+	2	5	9	8	1
RET EI	X	X	2	6	4	1	X	X	2	6	3	9
LAT	+	0	1	6	6	0	+	0	0	9	6	8
LONG	-	1	6	2	4	3	-	1	5	2	0	7

REMARKS: ① GMT UNDOCKING 150:01:12 ② ROLL RIGHT 55°/EMS ENTRY ③ GDC ALIGN TW SETTINGS FOR GDC ALIGNMENT TO ATM (G/3-19) ④ IF NOT IN SOLAR INERTIAL: ① GDC ALIGN WILL BE DONE DURING TIG-3 MIN HORIZ CK. ③ AFTER UNDOCKING TRACK AFT HORIZ ON 13° WINDOW MK (HDS DN)

REMARKS: ① GMT UNDOCKING 151:00:31 ② ROLL RIGHT
55°/EMS ENTRY ③ GDC ALIGN TW SETTING FOR GDC
ALIGNMENT TO ATM (G/3-19) ④ IF NOT IN SOLAR
INERTIAL; ⑤ GDC ALIGN WILL BE DONE DURING TIG
-3MIN HORIZ CK. ⑥ AFTER UNDOCKING TRACK
AFT HORIZ ON 13° WINDOW MK (HDS DN)

BLOCK DATA

GMT	TIG DAY	X	X	X	1	5	2	X	X	X			
	HR	X	X	X	X	0	0	X	X	X	X		
	MIN	X	X	X	X	0	2	X	X	X	X		
	SEC	X	X	X	X	3	2	X	X	X	X		
	ΔVC	X	X	4	3	1	6	X	X				
	BT	X	X	0	0	1	9	X	X				
	GDC ALIGN R	X	X	X	1	9	8	X	X	X			
	TW P	X	X	X	0	3	7	X	X	X			
	SETTINGS Y	X	X	X	0	0	0	X	X	X			
BRN ATT GDC FDAI		R, P, Y						0°, 180°, 0°					
	RET .05G	X	X	2	9	1	0	X	X				
	RTOGO .05G	+	1	1	8	9	0	+					
	VIO .05G	+	2	5	9	9	6	+					
	RET EI	X	X	2	6	3	9	X	X				
	LAT	+	0	3	1	8	3		0				
	LONG	-	1	7	3	0	0						

REMARKS: ① GMT UNDOCKING 151:22:16 ② ROLL LEFT

55°/EMS ENTRY ③ GDC Align TW SETTINGS FOR GDC

Alignment to ATM (Page G/3-19) ④ IF NOT IN SOLAR INERTIAL:

A) GDC Align will be done during Tig-3 MIN Horiz CK.

B) After undocking track AFT HORIE ON 15° Window

MR (HDS DN)

REMARKS:

DATE 5/17/73

DATE 5/17/73

BLOCK DATA

TIG DAY	X	X	X				X	X	X			
HR	X	X	X	X			X	X	X	X		
MIN	X	X	X	X			X	X	X	X		
SEC	X	X	X	X			X	X	X	X		
Δ VC	X	X					X	X				
BT	X	X					X	X				
GDC ALIGN R	X	X	X				X	X	X			
TW P	X	X	X				X	X	X			
SETTINGS Y	X	X	X				X	X	X			
BRN ATT GDC FDAI	R, P, Y						0°, 180°, 0°					
RET .05G	X	X					X	X				
RTOGO .05G	+						+					
VIO .05G	+						+					
RET EI	X	X					X	X				
LAT		0						0				
LONG												

REMARKS: _____

REMARKS: _____

BLOCK DATA

TIG DAY	X	X	X				X	X	X			
HR	X	X	X	X			X	X	X	X		
MIN	X	X	X	X			X	X	X	X		
SEC	X	X	X	X			X	X	X	X		
ΔVC	X	X					X	X				
BT	X	X					X	X				
GDC ALIGN R	X	X	X				X	X	X			
TW P	X	X	X				X	X	X			
SETTINGS Y	X	X	X				X	X	X			
BRN ATT GDC FDAI	R, P, Y						0°, 180°, 0°					
RET .05G	X	X					X	X				
RTOGO .05G	+						+					
VIO .05G	+						+					
RET EI	X	X					X	X				
LAT		0						0				
LONG												

REMARKS: _____

REMARKS: _____

DATE 5/17/73

BLOCK DATA

TIG DAY	X	X	X				X	X	X			
HR	X	X	X	X			X	X	X	X		
MIN	X	X	X	X			X	X	X	X		
SEC	X	X	X	X			X	X	X	X		
ΔVC	X	X					X	X				
BT	X	X					X	X				
GDC ALIGN R	X	X	X				X	X	X			
TW P	X	X	X				X	X	X			
SETTINGS Y	X	X	X				X	X	X			
BRN ATT GDC FDAI	R, P, Y						0°, 180°, 0°					
RET .05G	X	X					X	X				
RTOGO .05G	+						+					
VIO .05G	+						+					
RET EI	X	X					X	X				
LAT		0						0				
LONG												

REMARKS: _____

REMARKS: _____

DATE 5/17/73

BLOCK DATA

TIG DAY

HR

MIN

SEC

ΔVC

BT

GDC ALIGN R

TW P

SETTINGS Y

BRN ATT GDC FDAI

RET .05G

RTOGO .05G

VIO .05G

RET EI

LAT

LONG

X	X	X				X	X	X			
X	X	X	X			X	X	X	X		
X	X	X	X			X	X	X	X		
X	X	X	X			X	X	X	X		
X	X					X	X				
X	X					X	X				
X	X	X				X	X	X			
X	X	X				X	X	X			
X	X	X				X	X	X			
R, P, Y						0°, 180°, 0°					
X	X					X	X				
+						+					
+						+					
X	X					X	X				
0						0					

REMARKS:

REMARKS:

DATE 5/17/73

BLOCK DATA

TIG DAY	X	X	X				X	X	X			
HR	X	X	X	X			X	X	X	X		
MIN	X	X	X	X			X	X	X	X		
SEC	X	X	X	X			X	X	X	X		
ΔVC	X	X					X	X				
BT	X	X					X	X				
GDC ALIGN R	X	X	X				X	X	X			
TW P	X	X	X				X	X	X			
SETTINGS Y	X	X	X				X	X	X			
BRN ATT GDC FDAI	R, P, Y						0°, 180°, 0°					
RET .05G	X	X					X	X				
RTOGO .05G	+						+					
VIO .05G	+						+					
RET EI	X	X					X	X				
LAT		0						0				
LONG												

REMARKS:

REMARKS:

DATE 5/17/73

BLOCK DATA

TIG DAY

HR

MIN

SEC

ΔVC

BT

GDC ALIGN R

TW P

SETTINGS Y

BRN ATT GDC FDAI

RET .05G

RTOGO .05G

VIO .05G

RET EI

LAT

LONG

X	X	X				X	X	X			
X	X	X	X			X	X	X	X		
X	X	X	X			X	X	X	X		
X	X	X	X			X	X	X	X		
X	X					X	X				
X	X					X	X				
X	X	X				X	X	X			
X	X	X				X	X	X			
X	X	X				X	X	X			
R, P, Y						0°, 180°, 0°					
X	X					X	X				
+						+					
+						+					
X	X					X	X				
	0						0				

REMARKS:

REMARKS:

DATE 5/17/73